



FORM III TOPICS

1. a) i) What is meant by the term binomial nomenclature?

- scientific system of naming organisms using the generic(genus) and specific (species) names

ii) State briefly the general principles of classification of living organisms

- scientific names must be in Latin or should be latinised
- family names are formed by adding the suffix “idea” to the stem of the genus e.g. the genus Rana become Ranaidea
- generic names should be a single unique name

b) State the main characteristics of the five kingdoms of organisms

i. Monera

- e.g. bacteria
- unicellular (single celled)
- prokaryotic (genetic material not surrounded by membrane)
- cell wall without cellulose
- lack most organelles
- small in size (microscopic)

ii. Protista(protoctista)

- single celled(unicellular)
- eukaryotic (most cell organelles present)
- when cell walls are present have no cellulose
- e.g. protozoa and algae
- usually microscopic

iii. Fungi

- have hyphae (which form mycelia)
- absence of chlorophyll
- have rhizoids (lack roots, leaves, stem)
- have spore forming structures (sporangia)
- e.g. mucor, rhizopus

iv. Plantae

- most are green/contain chlorophyll
- autotrophic/feed by photosynthesis
- cells have cellulose cell walls
- respond slowly to stimuli (tropism)
- lack locomotion (are stationary)
- indefinite growth (at meristems)
- lack specialized excretory structures

v. Animalia

- cells do not have cell walls
- most carry out locomotion

- heterotrophic
- fast response to stimuli (tactic)
- have specialized excretory structures

c) Describe the economic importance of:

i. Fungi

- some cause decay to our food
- some cause diseases to humans and animals e.g. ringworms
- may be used as food e.g. mushrooms, yeast
- some are used in production of antibiotics e.g. penicillin, chloromycin, streptomycin
- yeast is used in brewing industry, baking and source of vitamin B
- many cause diseases to our crops e.g. late blight
- important in recycling nutrients in soil since they cause decay of organic matter
- mycorrhizal association in forest development may help in water intake/absorption
- help in nitrogen fixation

Bacteria

- are useful in the manufacture of antibiotics
- silage formation,
- fermentation of cheese, butter, milk yoghurt
- curing of tea, tobacco and retting flax
- formation of vitamin B12 and K
- enzymes such as amylase and invertase
- hormones such as insulin
- vinegar, acetic acid, lactic acid, citric acid
- in septic tanks and modern sewage works make use of bacteria
- biogas production
- saprophytic bacteria are used in compost decomposition or cause decay
- symbiotic bacteria are used in compost decomposition or cause decay
- symbiotic bacteria in herbivores/ruminants help in digestion
- some diseases in animals/humans and plants are caused by bacteria
- many bacteria cause food spoilage/decay
- nitrifying and nitrogen fixing bacteria increase soil fertility/make nitrates available
- denitrifying bacteria reduce soil fertility/convert nitrates into nitrogen/reduce nitrates

d) State the main characteristics of the following division of kingdom plantae



i. Bryophyte

- e.g. mosses and liverworts
- presence of rhizoids
- lack of vascular tissues (lack phloem and xylem)
- body parts not differentiated into root, stem, leaves
- capsule or seta
- gametophyte generation dominant.

ii. Pteridophyta

- e.g. ferns
- has true roots, stems and leaves
- found with sori on under-surface
- vascular tissues present
- sporophyte generation is dominant

iii. Spermatophyte

- photosynthetic
- well differentiated into roots, stems and leaves
- well developed vascular system
- seed bearing plants

e) Name sub-divisions of spermatophyte and state the characteristics of each class

i. Gymnospermae (cornifers)

- naked seeds (exposed)
- are all woody trees
- reproduce by means of cones
- show xerophytic characteristics
- xylem have tracheids but lack vessels
- phloem lacks companion cells
- single fertilization
- pollen lands directly on ovules

ii. Angiospermae (flowering plants)

- reproduce by flowers
- seeds enclosed (in fruits)
- flowers bisexual hence double fertilization
- herbaceous
- pollen grains land on stigma of pistil
- xylem contains vessels
- phloem contains companion cells

- ovules contained in ovary

iii. Name the classes and state characteristics of angiospermae

Dicotyledonae

- two seed leaves
- network venation of leaves
- regularly arranged vascular bundles
- tap root system
- broad leaves
- secondary growth occurs

Monocotyledonae

- one seed leaf
- parallel venation of leaves
- irregularly arranged vascular bundles
- fibrous root system
- narrow leaves
- sheath like leaf stalk (petiole)
- no secondary growth

iv) State the importance of plants

- balancing carbon IV oxide and oxygen in the atmosphere during photosynthesis and respiration
- influence water cycle
- reduce soil erosion by bind soil particles together
- useful products e.g. food, medicine, timber, paper and clothing
- habitat (e.g. forests and grassland) for animals which may also be tourist attraction
- earn money from sales of products
- aesthetic value/beauty e.g. flowers, shade/shelter, live fences, windbreaks
- Some are harmful e.g. poisons, weeds, injurious (stinging nettles, thorns), water hyacinth.

f) i) Give the general characteristics of phylum arthropoda

- jointed appendages
- presence of exoskeleton
- triploblastic and coelomate
- segmented body



- bilateral symmetry (similar halves)
- ii. **State the characteristics of the following classes of arthropoda**

Diplopoda

- the millipedes
- two pairs of legs per segment
- many segments
- terrestrial habitat
- body cylindrical and long
- herbivorous
- one pair of antennae

Chilopoda

- the centipedes
- one pair of legs per segment
- many segments
- terrestrial habitat
- body long and ventro-dorsally flattened
- carnivorous
- last pair of legs pointing backwards with poison claws called maxillipedes
- one pair of antennae

Insecta

- three body parts i.e. head thorax, abdomen
- six legs/three pairs of legs
- a pair of compound eyes
- presence of wings
- a pair of antennae

Crustacean

- two body parts
- segmented body
- have pincers (modified legs) to catch prey
- have hard exoskeleton
- a pair of compound eyes

Arachnida

- body divided into two parts(abdomen and cephalothorax)
- simple eyes
- eight legs (four pairs of legs)

iii) State the economic importance of insects

Beneficial effects

- food supply
- important in food chains
- pollinators
- biological control of pests and other organisms
- aesthetic value
- contribute to decomposition e.g. litter feeders like beetles

Harmful effects

- pests
- vectors
- dirt and disease carriers
- injurious e.g. stings and bites

g) i) State the general characteristics of chordate

- notochord
- dorsal slits (pharyngeal cleft during development)
- bilateral symmetry
- triploblastic (three layer body-ectoderm, mesoderm and endoderm)
- clear cut head formation
- multilayered epidermis
- post anal tail
- closed circulatory system
- segmented muscle blocks(myotomes)
- single pair of gonads



- hollow bones
- oviparous (lay eggs)
- mouths modified into beaks

Give the characteristics of the following classes of chordate

Pisces

- presence of fins for locomotion
- two chambered heart
- presence of overlapping scales
- presence of gills or operculum for gaseous exchange
- presence of lateral line for protection
- streamlined body
- poikilothermic (body temperature varies with that of environment)

Amphibian

- partially live in fresh water and partially on land
- poikilothermic
- pentadactylous with two pairs of limbs
- webbed feet for locomotion in water
- body streamlined
- heart is three chambered
- moist skin for gaseous exchange

Reptilia

- scales on body
- poikilothermic
- homodont teeth except tortoise and turtle
- all have limbs except snakes
- skin is dry
- oviparous (lay eggs)
- no pinna (external ear)
- three chambered heart (crocodile has four chambers)
- skin not glandular
- no mammary glands

Aves

- the birds
- homoeothermic (constant body temperature)
- four chambered heart
- streamlined body for locomotion in air
- skin dry and covered by feathers
- scales on legs

Mammalian

- hair on the body
- homoeothermic
- viviparous (give birth to live young) except a few
- have mammary glands
- glandular skin e.g. sweat glands, sebaceous glands
- four chambered heart
- pinna (external ear)
- two pairs of pentadactyl limbs
- presence of diaphragm
- have salivary glands

a) i) What is a dichotomous key?

- A biological device (tool) which enables one to identify an organism by progressively opting between two alternative observable characteristics

i. State the necessity of using a dichotomous key

- used to identify organisms quickly and accurately
- by following the statements in the key we are able to identify each organism on the basis of a characteristic which is not to be found in other specimens

ii. List the rules followed in constructing a dichotomous key



- use observable characteristics only
- start with major characteristics, placing organisms into two groups at each stage
- use a single characteristics at a time
- use contrasting characteristics at each stage e.g 1(a) short, 1(b) tall
- avoid repeating the same characteristics

iv) Describe the procedure of using a dichotomous key. Make a list of major features of the characteristics to be identified

- look at the features of similarities
- look at the features of differences between the organisms
- we can then be able to identify the organisms by distinguishing one from another
- the key uses a method of elimination by following statements that are correct only for the organism

- | | |
|---|--------------|
| 1 a) leaf broad | go to 2 |
| b) leaf narrow | Araucaria |
| 2 a) leaf parallel veined | Cynodon |
| b) leaf net-veined | go to 3 |
| 3 a) leaf with one lobe (simple) | go to 4 |
| b) leaf with many lobes (compound) | Grevellea |
| 4 a) leaf fleshy | Kalanchoa |
| b) leaf not fleshy | go to 5 |
| 5 a) leaf petiole modified to form sheath | go to 6 |
| b) leaf petiole not modified to form sheath | Brassica |
| 6 a) leaf purple | Tradescantia |
| b) leaf green | Commelina |

steps – 1a, 2b, 3a, 4b, 5b

Identify – Brassica

v) You have been provided with four animals labeled K (mature adult housefly), L (mature adult grasshopper, M(maize flour beetle) and N(worker termite) use the dichotomous key below to identify the specimens. Write down in the correct order, the steps (number and letter) in the key that you followed to arrive at your answer.

Dichotomous key

- | | |
|-------------------------|---------|
| 1 a) animal with wings | go to 2 |
| b) animal without wings | go to 7 |

iv You are provided with a specimen kale leaf. Use the dichotomous key below to identify the taxonomic group to which the specimen belongs. Show the steps (number and letter) in the key that you followed to arrive at the identify of the specimen



2 a) with two pairs of wings
go to 3

b) with one pair of wings
Diptera

3 a) with membranous wings
go to 4

b) hind pair of membranous wings
go to 6

4 a) with long abdomen
Odontata

b) medium sized abdomen
go to 5

5 a) wings with coloured scales
Lepidoptera

b) wings without scales
hymenoptera

6 a) forewings hard and shell-like
coleopteran

b) forewings hard but not shell-like
Orthoptera

7 a) body horizontally flattened

Isoptera

b) body laterally flattened
Symphonoptera

Identify the orders of the various specimens as per the table below

Specimen	Order	Steps followed
K-housefly	Diptera	1a,2b
L- grasshopper	Orthoptera	1a, 2a, 3b, 6b
M- beetle	Coleoptera	1a, 2a, 3b, 6a
M-termite	Isoptera	1b, 7a

2 a) Define the following ecological terms

i. Ecology

- study of the interrelationships between organisms and their environment

ii. Environment

- surrounding of the organism i.e. biotic or a biotic factors

iii. Habitat

- A specific locality (home) of a living organism with a set of factors (conditions) in which an organism lives.

iv. Ecological niche

- Role of an organism in its habitat e.g. feeding relationship

v. Population

- Number (group) of organisms of a species occupying a given habitat

vi. Community

- Refers to different species of (plants and animals) organisms in a given habitat (area) co-existing or interacting (living) with each other and the environment in which they live

vii. Ecosystem

- A community of organisms interacting with one another and the environment in which they live

viii. Biosphere

- The earth and its atmosphere where living organisms are found

ix. Autecology

- Study of a single (individual) species of plants or animals within a community, ecosystem, habitat or environment.

x. Synecology

- Study of natural communities (plants and animals) or populations interacting within an ecosystem.



xi. Carrying capacity

- maximum number of organisms an area can support without being depleted

xii. Biome

- geographical area with particular climatic conditions and flora and fauna
- it constitutes many ecosystems

xiii. Biomass

- dry weight (mass) of a living organism in a given area
- units of measurement are kg/m²/year

b) i) What are abiotic factors?

- non-living components of the ecosystem

ii) Explain how abiotic factors affect living organisms

Wind

- this influences rate of water evaporation from organisms
- therefore it affects distribution of organisms e.g. wind increases rate of transpiration and evaporation of water from the soil
- wind is an agent of soil erosion, may break and uproot trees
- may aid in the formation of sand dunes which can form habitats for some desert plants
- wind disperses fruits, seeds, spores
- wind forms waves in lakes and oceans which enhances aeration of water which replenishes oxygen concentration necessary for life
- wind is an agent of pollination

Temperature

- influences rate of enzyme action in photosynthesis and other metabolic reactions in plants and animals
- organisms function within a narrow range of temperature
- it affects distribution of organisms
- changes in temperature affect rate of photosynthesis and biochemical reactions e.g. metabolism and enzyme reaction
- temperature increases rate of transpiration

Light

- needed by green plants and photosynthetic bacteria which are primary producers
- animals depend on plants directly or indirectly for food
- main source of light is the sun
- light is necessary for synthesis of vitamin D in certain animals
- some plants need light for flowering
- seeds like lettuce need light for germination

Humidity

- amount of water vapour held by the air
- affects the rate at which water is lost from organisms body by evaporation and stomatal transpiration
- when humidity is low the rate of transpiration increases
- humidity influences distribution of organisms

PH

- each plant requires a specific PH in which to grow (acidic, neutral or alkaline)
- pH affects enzyme reaction in metabolism

Salinity

- some ions are needed for plant and animal nutrition
- osmoregulation in plants and animals is affected by salinity

Topography

- altitude affects light, atmospheric pressure and light
- Slope influences surface runoff, wind erosion, etc.
- mountains affect distribution of organisms which differs in leeward side and windward side
- mountains affect distribution of organisms which differ on lowlands and on highlands
- mountains also form physical barriers to migration of organism and may cause isolation of species
- background may offer camouflage to some organisms hence protection from enemies

Rainfall (water) or precipitation

- amount and distribution of rainfall affect vegetation type
- this consequently affects distribution of animals e.g. polar region water frozen hence only well adapted organisms survive
- fewer organisms found in deserts where rainfall is less
- Water is required for seed germination, raw material for photosynthesis, solvent for mineral salts. Provides turgidity for plant support, medium for transport, disperses fruits, seeds and spores

Pressure

- the weight atmosphere exerts upon the earth
- varies with altitude the higher the altitude the less the pressure
- this variation implies change in density which directly means less oxygen for respiration and less carbon iv oxide for photosynthesis and this affects distribution of organisms

Mineral salts (trace elements)



- these affect distribution of plants in the soil
- plants thrive best where elements are available
- Plants living in soil deficient in a particular element must have special methods of obtaining it.
- They harbor nitrogen fixing bacteria and others have carnivorous habit
- Plant distribution influences animal distribution

c) i) What are biotic factors?

- refers to living organisms in an area
- biotic environment of an organism constitutes all organisms around it, which it relates or interacts with in various ways

ii) Give examples of biotic factors affecting ecosystems

- feeding relationships
- predation
- competition
- diseases and pests
- human activities

d) Discuss how the various biotic factors affect living organisms

i. Competition

- organisms compete with one another for food, light, water, mates and shelter
- organisms must live together for competition for available resources
- those which cannot cope either structurally or behaviorally will migrate or die
- those remaining, due to better adaptations will increase in population
- competition between members of the same species is called intra-specific competition e.g. for mates
- Competition between members of different species is inter specific competition e.g. for food and space.

ii. Predation

- this is predator-prey relationship
- predator feeds on prey hence both control the other's population
- Distribution of predator and prey is important as predator cannot survive without prey
- If there is no predator the prey will increase in population beyond carrying capacity hence die due to environment depletion

iii. Parasitism

- an association where an organism lives in or on another living organism obtaining food (and other benefits) from it, causing harm to it (without necessarily killing it)
- parasites may kill host

- they deprive host of food
- make host weak by introducing diseases
- make reproductive ability of host low hence host becomes susceptible to predation

iv. Diseases and parasites

- make organisms weak and susceptible to predation
- kill organisms and reduce their population

v. Symbiotic

- and association of organisms of different species where both benefit from the association i.e. there is mutual benefit

vi. Human activities

- these are human factors which have an influence on the biosphere
- examples are road construction, industrialization, deforestation, agriculture, pollution, poaching, fishing conservation, population control
- affect ecosystem and balance of nature

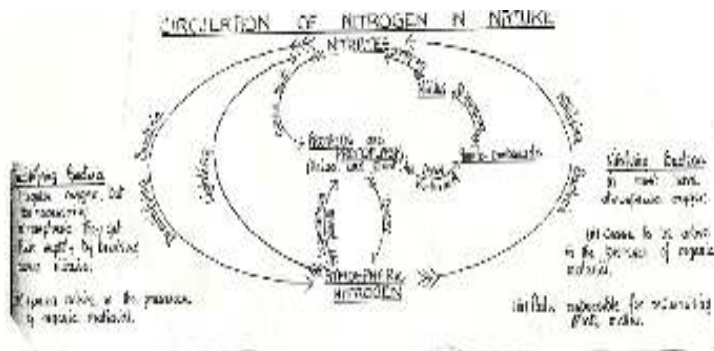
Saprophytism

- saprophytes are organisms which obtain organic matter in solution from dead and decaying tissues of plants and animals
- they include saprophytic bacteria and fungi
- they make available carbon, nitrogen and other elements from dead to living organisms
- they are useful in recycling nutrients in nature

e)i) What is nitrogen cycle?

- The process by which nitrogen in the air is made available plants and animals and eventually returns to the air.

ii) Draw a simplified diagram representing the nitrogen cycle



pseudomonas and theobacillus bacteria generally called denitrifying bacteria

iii. Nitrogen in the atmosphere cannot be directly utilized by plants. State two ways by which this nitrogen is made available for plant use

- fixation by microorganisms (Rhizobium, Azotobacter)
- fixation by electrical discharge in atmosphere i.e. conversion by thunderstorm or lightning

f) i) Describe how energy flows from the sun through the various trophic levels in an ecosystem

iii) Describe the nitrogen cycle

- during thunderstorms/lightning nitrogen gas combines with oxygen to form nitrogen oxides
- nitrogen oxides dissolve in water to form nitric acid
- acid is deposited in the soil by rain
- nitric acid combines with chemical substances to form nitrates or nitric acid dissociates to form nitrates which are absorbed by plants
- symbiotic bacteria (Rhizobium) which are found in root nodules of leguminous plants fix free nitrogen to nitrates
- free living bacteria (clostridium and Azotobacter) fix nitrogen to nitrates
- nostoc algae (Anabaema chlorella) fix nitrogen to nitrates
- plants use nitrates to form plant proteins
- animals feed on plants and convert plant proteins into animal proteins
- plants and animals die and are decomposed by putrefying bacteria, fungi(saprophytes)
- decomposing plants, animals and nitrogenous wastes release ammonia which is converted to nitrites by Nitrosomonas and nitrococcus bacteria

- energy from the sun is trapped by green plants during photosynthesis, producing chemical energy9food or carbohydrates
- green plants are producers and occupy the first trophic level
- green plants are eaten by herbivores called primary producers as they occupy the second trophic level
- herbivores are eaten by carnivores, secondary consumers, which occupy the third trophic level
- when organisms 9plant and animals) die, fungi and bacteria which are saprophytic organisms feed on them thus causing them to decompose into simple substances e.g. mineral salts
- these organisms are called decomposers and detritivores
- decomposer feed on dead organic matter hence cause decomposition and decay which releases nutrients for plants, linking biotic and a biotic components
- at all levels energy is lost through respiration

Give the reasons for loss of energy from one trophic level to another in a food chain

- nitrites are converted to nitrates by nitrobacter bacteria
- nitrates in the soil can be converted to free nitrogen (denitrification) by some fungi,

- insufficient utilization of food resources(wastage) e.g by defalcation
- through respiration
- through excretion e.g. urination and sweating

Why are green plants referred to as primary producers in an ecosystem?



- They utilize the energy from the sun to manufacture food for themselves and for subsequent trophic level (consumers) and other organisms

vi. Explain the following terms giving suitable examples

Food chain

- a nutritional sequence between producers and consumers through which energy flows in a straight line i.e. linear representation of feeding relationship between different organisms in an ecosystem
- if one consumer or the producer is removed the food chain is broken
- arrow points to the direction of energy flow e.g. green plant herbivore carnivore decomposer

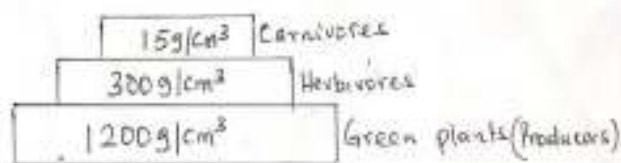
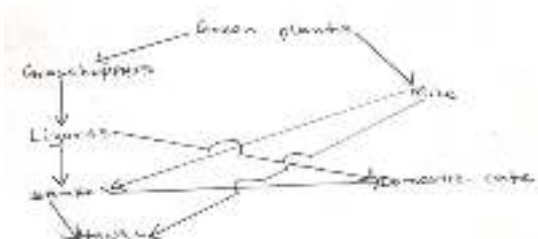
Green plants → herbivores → Carnivore

- this is a diagrammatic representation of numbers of organisms at each trophic level in a food chain
- usually there are more producers than consumers
- hence producers herbivores carnivore
- the reason for the pyramid is because herbivores feed on many plants (producers) as carnivores feed on many herbivores
- sometimes this may not be true e.g. when many caterpillars feed on one tree or parasites on a herbivore
- this gives an inverted pyramid of numbers

Pyramid of biomass

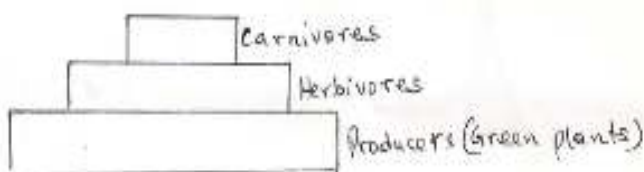
Food web

- complex feeding relationship where a consumer feeds on more than one type of food while several herbivores feed on one type of plant
- it is an interrelationship of many food chains
- consumers are usually fewer than producers to ensure survival of both



- refers to diagrammatic representation total dry weight of organisms at different trophic levels in a food chain
- producers have greater biomass than any level of consumers progressively
- size of organisms in successive e trophic levels increases
- amount of individuals decreases in successive levels

Pyramid of numbers





Account for the decrease of biomass in the successive trophic levels

- fixed energy which supports living matter decreases at each successive trophic level since energy is lost by respiration and indigestible (unconverted) materials hence less biomass supported at each level

h) i) Describe the three characteristics of a population growth

- increase in numbers
- decrease in numbers/growth rate
- change in numbers

Dispersion

- spread or distribution of organisms in a habitat

Density

- the number of individuals per unit area

ii) Explain how the following methods are used to estimate population of organisms

quadrat method

- identify the study area
- throw or mark out the quadrat in the area of study at random
- identify or label the various species of plants in the quadrat
- count plants of each species
- record the numbers
- repeat the process
- work out the average per quadrat for each species
- calculate the total number of different species in the area or calculate the population for the total area of habitat

Line transect

- a string is stretched along an identified area
- all plants touching the string are counted

Belt transect

- preliminary study of the study area to estimate size or make a sketch map
- two parallel lines (strings or ropes) running for a determined distance and width
- count the number of organisms in the transect
- calculate the area covered by the transect
- calculate the number of organisms being investigated per unit area
- repeat this process at least three times in other parts of the study area
- find the mean number of organisms per unit area from all the belt transects
- from this figure calculate the total population of the desired organisms in the study area.

Capture-recapture method

- e.g. grasshoppers or fish
- capture the grasshoppers
- count and mark using permanent ink
- record
- release and allow time
- recapture and count the marked and unmarked
- total population is equal to the number of marked and unmarked grasshoppers in the second sample multiplied by the number of marked grasshoppers in the first sample divided by number of grasshoppers marked in the second sample that were recaptured

2. a) Describe the adaptations of plants to various habitats

i. Xerophytes

- grow in areas with scarcity of water
- roots grow deeply and extensively (widely spread) to ensure access to water
- thick succulent stems, roots and leaves for water storage



- photosynthetic stems take place of leaves which would lose a lot of water
- Leaves are needle-like (reduced to spines), scaly, have sunken stomata. Some have curled (rolled) leaves. Some have thick waxy cuticle, reduced number of stomata to reduce water loss by transpiration
- some shed leaves during dry season to reduce water loss
- presence of thorns for protection
- short life cycle to ensure survival
- reversed stomatal rhythm

ii. Hyrophytes

- grow in places with plenty of water(waterlogged)
- aerenchyma a tissue (airspaces) and large intercellular spaces and long fibrous roots for buoyancy (floating in water)
- poorly developed support tissues (sclerenchyma) because water provides the necessary support
- upper epidermis of leaves have more stomata than lower epidermis for gaseous exchange or for increased rate of transpiration
- poorly developed conducting tissues (xylem and phloem) because plants obtain water by diffusion

iii. Mesophytes

- grow in well watered soils (common plants)
- no special adaptations, but depending on particular habitat, may have some adaptations
- in forests they grow fast, tall to capture light. Have climbers while some are adapted to carry out photosynthesis in low light intensities (those that form undergrowth)
- in places with adequate water they form broad leaves, thin cuticle and many stomata on both leaf surfaces
- in drier regions they possess more stomata on the lower leaf surface and are deep rooted
- some are shallow rooted and develop buttress and prop roots for support
- some have waxy or glossy surface to reflect sun rays and drip off rain water

iv. Halophytes

- plants that grow in very salty soil where the salt concentration is higher than that in the plant
- have root cells which concentrate a lot of salts in them and enable them to take in water by osmosis
- succulent roots to store water
- have pneumatophores (breathing roots) to take in oxygen
- some have buttress roots for support
- secrete excess salt by use of salt glands
- have large airspaces in leaves and stems for buoyancy and to store air
- capable of photosynthesis at low light intensities
- e.g. mangrove

b) i) What is pollution?

- any process which leads to adverse or harmful changes in the environment

ii) Explain the various human activities that have caused pollution

Causes and effects of air pollution

- sulphur iv oxide, hydrogen sulphide, chlorine, oxides of nitrogen produced by industries, sewage, decomposing organic matter and fumes affect gaseous exchange, makes acid rain and damage plant leaves
- aerosols, herbicides, insecticides (agrochemicals), paint spays, acaricides and CFC's sprayed to control diseases, pests and weeds affect respiratory organs of animals. The chemicals are residual and persistent (not easily broken down) and bring depletion of the ozone layer
- smoke and fumes produced in areas with heavy industries, motor vehicles, fires which burn fuel, oil, wood and coal cause carbon ii oxide, poisoning affect respiratory systems and affect visibility
- particles in smoke and fumes settle on leaves and stop photosynthesis
- carbon iv oxide causes green house effect which causes temperature inversion as a result of heating the lower layers of atmosphere
- sound and noise produced incessantly by machines, aeroplanes and heavy vehicles affect hearing in animals
- dust from cement factories, quarries, dust roads settles on leaves limiting photosynthesis



- removal of vegetation interferes with carbon cycle
- radio-active emissions from nuclear reactors, mines and bombs cause cancer, mutations and death.

Control of air pollution

- use of lead free petrol in motor vehicles, air craft, aeroplanes and petroleum engines
- uses of smokeless fuels and electricity
- filtration, dissolution and use of chemicals to remove harmful gases
- factories should be erected far away from residential areas
- use of tall chimneys
- reduce volume or intensity of sound e.g. by use of ear muffs
- concords should fly at higher altitudes and aeroplanes to fly high up

State the causes, effects and methods of controlling and prop roots for support water pollution

Causes and effects

- agrochemicals e.g. fertilizers cause eutrophication leading to increase in animal population
- Silting makes water surfaces shallow and silt clogs stomata and gills of fish reducing rates of photosynthesis and gaseous exchange. It also leads to reduction of algae which causes reduction of consumers i.e. animal population
- industrial and domestic wastes contain toxic materials which kill producers and other organism while oily substances in wastes may clog gills of fish and may change pH of water oxygen solubility is also reduced by oily surfaces

- Untreated sewage and effluents where decomposition of organic matter in sewage reduces oxygen supply and sewage provides food for bacteria increasing their population and demand for oxygen thus depriving fish of oxygen.
- Human faeces causes eutrophication, carbon IV oxide produced by decomposition of faecal matter changes pH of water interferes with photosynthesis and may clog fish gills or block light penetration which interferes with producers thereby decreasing productivity.
- Dumping of chemicals from industries with toxic pollutants which kill organisms
- Spillage of oil and chemicals block oxygen and kill organisms
- Discharge of water from industries into water body where high temperatures reduce amount of oxygen in the water causing organism to suffocate and die
- Untreated sewage may lead to outbreak of epidemics

Control of water pollution

-pollution caused by domestic effluents may be controlled by treating domestic waste, using biotechnology, banning the use of phosphate-based detergents, using plastic pipes instead of those made from lead, recycling gabbage, using biodegradable detergents.

Pollution caused by industrial waste may be controlled by treating/cooling industrial waste, carrying out environmental impact assessment before establishing industries

Oil spillage may be controlled by cleaning spilled oil biotechnology and penalizing the industry individual or companies which cause oil spills/water pollution

Pollution caused by agrochemicals may be controlled by using mechanical control of weeds, biological control of weeds and pests, biodegradable organic fertilizer herbicides, insecticides pesticides, organic farming educate farmers on the use of correct amount of agrochemicals

- silting may be controlled by appropriate farming practices, contour farming, reafforestation, building gabions and terracing



Control of soil pollution

- use of organic farming techniques
- biological control of pests, diseases, parasites
- recycling of non-degradable containers or burying them safely after use
- controlled burning of garbage
- treatment of human and industrial waste for safe disposal
- avoid spilling chemicals and oil when used

v) Define biological control give suitable examples

- using a living organism to regulate, control or reduce the population of another organism e.g beetles to feed on water hyacinth, fish to feed on mosquito larvae.

vi) What is eutrophication?

- enrichment of water bodies with nutrients due to discharge of sewage leading to rapid growth of surface plants

i) What are the effects of eutrophication?

- enrichment of water bodies with nutrients due to discharge of sewage leading to rapid growth of surface plants

vii) What are the effects of eutrophication?

- The plants block light from reaching plants underneath hence no photosynthesis

The plants die and decompose leading to lack of oxygen hence animals also die

iv) State the causes /effects and control methods of soil pollution

Causes and effects

- Air pollutants e.g. sulphur IV oxide fumes form sulphuric acid with rain water. The acid rain alters soil pH therefore affecting plants that cannot tolerate acidic soil
- most aerosols sprayed to control pests and diseases precipitate in the soil and are taken up by plants which make its concentration many times higher, increasing the toxicity in the plants which absorb them
- petroleum products due to spillage by oil tankers making it impossible for plant roots to obtain oxygen in oil saturated soils, therefore plants are killed
- agrochemicals and inorganic fertilizers contain heavy metals that are not used up by plants and eventually soil microorganisms cannot inhabit the soils
- organic matter slows down, life ceases and soil becomes exhausted
- community, household wastes and industrial wastes disposal is a major problem in big towns and cities. commodities packaged in metal tins, rubber, plastic containers, scrap metal, glass bottles, different types of paper are nuisance to the environment, rendering it useless for agricultural purposes



c) Describe the symptoms, mode of transmission and control of cholera, typhoid malaria and amoebic

dysentery in humans

Disease	Causative agent	Transmission	Symptoms	Control
Cholera	Vibrio cholerae(bacterium)	Contaminated food or water Spread by flies from faeces	Intestinal pain Diarrhea Vomiting Dehydration	Proper hygiene e.g. boiling drinking water Vaccination
Typhoid	Salmonella typhi (bacterium)	- contaminated food or water - spread by flies from faeces	- fever - rashes - diarrhea + blood from bowels	- proper sanitation - vaccination
Malaria	Plasmodium (protozoa)	- bite by infected female anopheles mosquito	- fever Joint pains Vomiting Headache Anaemia	- killing mosquito - killing the mosquito larvae - draining stagnant water - clearing bushes - treatment - sleep under mosquito net
Amoebic dysentery (amoebiasis)	Entamoeba histolytica (bacterium)	- contaminated food or water due to improper disposal of faeces	- intestinal pain Diarrhea Vomiting Dehydration	- sanitation - personal hygiene - cook food well - treatment using drugs



d) Discuss *Ascaris lumbricoides* under the following sub-headings

i. Mode of transmission

- through ingestion of contaminated food
- live in intestines

ii. Effects of parasite on the host

- inflammation of lungs
- pneumonia
- produce toxic substances
- intestinal obstruction

iii. Adaptations

- thick cuticle which protects it against digestion
- lays many eggs to ensure survival
- mouthparts for sucking partly digested food
- lack of elaborate alimentary canal
- tolerant to low oxygen concentration
- two hosts to ensure survival
- eggs have protective cover to ensure survival in adverse environments

iv. Control and prevention

- proper sanitation
- wash hand after defaecation and before eating

- larvae have a sucker for attachment on human skin which it digests
- larva has a tail which it swims with in search of host in water
- prolonged association between male and female to ensure that fertilization takes place
- adults can tolerate low oxygen concentration (in the animal tissues)
- adult worm secretes chemicals against antibodies
- larvae and eggs (have glands that) secrete lytic enzymes to soften the tissues that ease penetration
- larvae are encysted so as to survive adverse conditions

e) Discuss *Schistosoma* under the following sub-headings

i. Mode of transmission

- through contaminated water in swamps, etc

ii. Effects on host

- bleeding in lungs
- blood stained urine
- unthriftiness

iii. Adaptations

- has two hosts to increase chances of survival
- eggs have a hook like structure which raptures the walls of intestine or bladder
- lay large number of eggs to ensure survival

iv) Control and prevention

- proper use of toilet facilities
- boiling water before use
- avoid bathing/washing in infected water
- Use of molluscicides (chemicals that kill snails/biological control/clearing water weeds on which snails feed.
- Drainage of stagnant water
- Wearing gum/rubber boots

3. a) i) What is reproduction?



- process by which living organisms give rise to new members of their own species which resemble the parents

ii) Why is reproduction important?

- for continuity of species/ to ensure survival of species
- maintaining life of species
- replace dead individuals

iii) Name the types of reproduction

- sexual which involves fusion of male and female gametes
- asexual in which no gametes are involved but parts of a mature organism develops into new individuals

b) i) What is cell division?

- process by which cells are formed from pre-existing cells

ii) What are chromosomes?

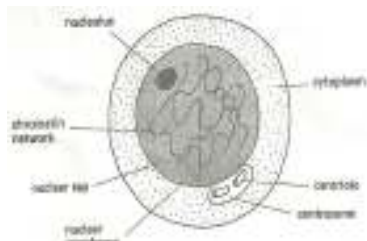
- Threadlike structures found in nucleus of a cell.
- The units called genes
- Genes are factors that cause inheritance or determine characteristics of offspring

c) i) What is mitosis?

- A type of cell division that occurs during growth leading to increase in number of cells
- all cells maintain the same chromosome constitution i.e. the diploid state

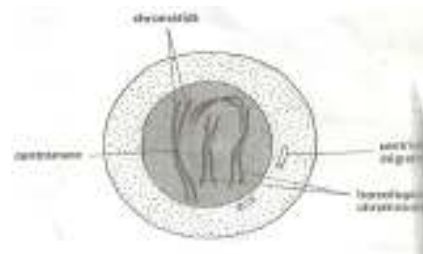
ii) Describe the five stages of mitosis

Interphase



- replication of organelles
- duplication of DNA
- production of energy (ATP) for cell division

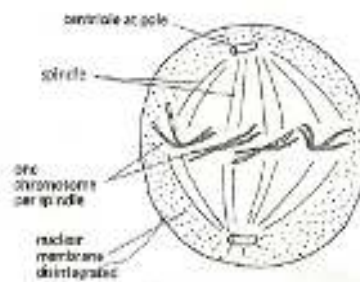
Prophase



- stage of dehydration
- chromosomes shorten and thicken
- chromosome replicates into two chromatids
- chromatids joined at centromere
- formation of spindle fibers

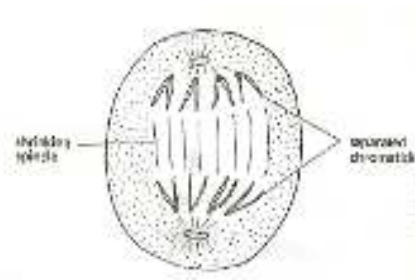
Metaphase

- chromosomes move to equator (early metaphase)
- chromosomes line up at the equator
- homologous chromosomes do not associate



Anaphase

- chromatids separate
- move to opposite ends (poles) of the cell



ii) State the significance of mitosis

- ensures each daughter cell has same number and kinds of chromosomes as daughter cells
- gives rise to new cells (responsible for growth)

d) i) What is meiosis?

- division of diploid cells to form gametes which are haploid

ii) State the significance of meiosis

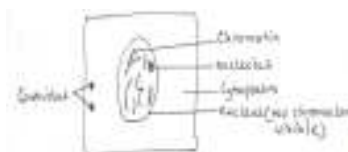
- gives rise to gametes
- source of variation

iii) Give a summary of the stages of meiosis

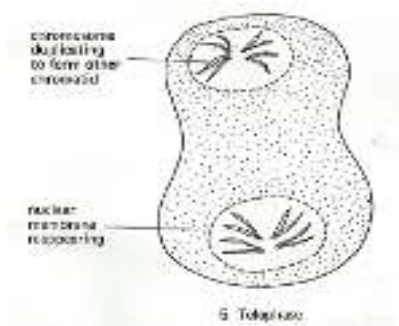
First meiotic division

Interphase

- cell is in non-dividing condition
- chromosomes appear threadlike



Telophase



Prophase I

- chromatic material shorten and thicken
- double stranded chromosomes appear (bivalent)
- double stranded chromosomes pair and twist round each other (synapsis)
- point of contact of chromosomes is called chiasma



- chromatids reach the poles
- formation of two daughter cells occurs i.e. cytoplasmic division



Metaphase I

- paired homologous chromosomes line up at the equator



- movement of chromosomes to equator



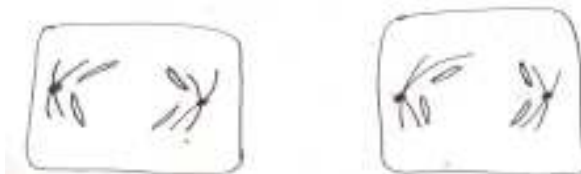
Anaphase I

- paired homologous chromosomes move to the poles



Anaphase II

-chromatids of each chromosome separate to the poles



Telophase I

- paired homologous chromosomes reach the poles
- two new nuclei are formed



Telophase II

- reach the poles
- four haploid daughter cells are formed



Second meiotic division

Prophase II

- chromosomes shorten, thicken and become visible,
- stage of dehydration



iv) Give the similarities between mitosis and meiosis

- both take part in cells
- both involve division (cell multiplication)

v) What are the differences between mitosis and meiosis?

Mitosis	Meiosis
- maintenance of chromosome number (diploid) - takes place in somatic	- reduction/halving of chromosomes (haploid) - occurs in reproductive cells/gonads/produce

Metaphase II



cells/growth • no crossing over/no variations • results into 2 daughter cells • no pairing/no synapsis/no bivalent formed • a one division process of four stages	s gametes • crossing over takes place/variation occurs • results into 4 daughter cells • there is pairing/synapsis/bivalent • a two division process of four stages each
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d) i) What is asexual reproduction

- formation of new individuals as a result of the fusion of two gametes
- fusion is called fertilization

ii) What is the significance of sexual reproduction in living organisms?

- leads to genetic variation e.g. cross breeding which gives rise to hybrids

iii) State the advantages of sexual reproduction

- genetic variation
- greater adaptability to environment by offspring
- few bad or good traits inherited/retained
- greater amount of dispersal is possible
- may result in stronger offspring

iv) Give the disadvantages of sexual reproduction

- less certainty in egg and sperm meeting
- low rate of survival
- sex-linked diseases easily transmitted

e) i) What is asexual reproduction?

- formation of new organisms without fusion of gametes
- occurs with only one parent
- parts of organism develop into new individual

ii) State the advantages of asexual reproduction

- retention of useful characteristics/genes/traits
- offspring establish faster/shorter life cycle

- better chances of survival because of suitable environment

iii) Give the disadvantages of asexual reproduction

- lack of genetic variation
- lowered resistance to disease
- loss of hybrid vigor
- competition for resources due to overcrowding

iv) Explain how reproduction occurs by the following methods of asexual reproduction

Sporulation

- formation of spores
- spores are small haploid cells produced by plants
- spores give rise to new haploid organisms
- includes moulds, ferns, bryophytes, pteridophytes

Budding

- where an outgrowth arises from a parent and drops off to develop into a new organisms
- hereditary material in the daughter cell and parent are exactly the same
- occurs in organisms such as hydra, jelly fish, sea anemones, yeast and some fungi

Binary fission

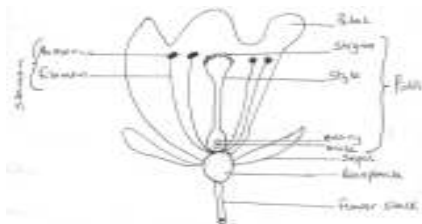
- a cell splits into two new cells of equal size
- each daughter cell grows into a new organism
- Occurs in organisms such as amoeba, euglena, paramecium, some fungi and bacteria.

f) i) What is a flower?

- this is the reproductive structure which bears the reproductive parts of a plant
- it produces seeds and fruits



ii) Draw a longitudinal section of a labeled diagram of a flower



- each carpel contains one or more ovules in an ovary
- style bearing a stigma extends from ovary
- ovary contains female gametes which when fertilized become seeds

iv) What is inflorescence?

- a group of flowers borne on the same branch (main stalk)

v) Explain the meaning of the following terms which describe flowers

iii) Give the functions of the parts of a flower

Receptacle

- expanded end of stalk which bears floral parts

Calyx

- consists of sepals
- usually green
- protect flower in bud

Corolla

- consist of petals
- often colored or scented to attract insects

Androecium

- male part of flower
- consist of stamens
- each stamen consists of an anther containing pollen sacs
- anther produces pollen grains which contain male gametes

Gynaecium

- female part of flower
- consists of one or more carpels

Hermaphrodite

- one with both stamen and carpel
- most flowers are hermaphrodite/bisexual

Unisexual

- have only one of carpel or stamen i.e. either male or female

Carpelate

- also called pistilate
- contains only carpels hence a female flower

Staminate

- also called male flower
- contains only stamens

Dioecious plants

- have pistilate and staminate flowers on different plants e.g. pawpaw

Monoecious plants

- have pistilate and staminate on one plant
- however, pistilate and staminate occur at different plants e.g. maize

Complete flower

- Has all four parts i.e. Calyx, corolla, androecium and gynoecium



Incomplete flower

- does not have all four parts
- at least one is missing

vi) Explain the meaning of the following types of ovary

Superior

- ovary occurs above other floral parts on the receptacle

Inferior (epigynous)

- other floral parts arise above ovary on the receptacle

g) i) What is pollination?

- transfer of pollen grains from anther of a stamen to stigma of a flower

ii) Explain the types of pollination

- self pollination takes place when mature pollen grains of a flower fall on the stigma of the same flower
- cross pollination takes place when pollen grains of a flower fall on the stigma of another flower of the same species

iii) State the advantages of pollination

- healthy offspring
- leads to variation
- greater chances of dispersal

iv) List the agents of pollination

- wind
- water
- insects

v) How are flowers adapted to wind and insect pollination?

Insect pollinated flowers (entomophilus)

- are scented to attract insects
- have sticky stigma for pollen grains to stick on
- are brightly coloured to attract insects
- presence of nectar to attract insects
- have nectar guides to guide insects to the nectarines
- have nectarines to secrete nectar
- stigma/ anthers located inside the flower/tubal/funnel shaped corolla to increase chances of contact by insects
- sticky/spiny/spiky pollen grains which stick on the body of insects and on stigma
- large/conspicuous flowers easily seen by/attract insects
- anthers firmly attached to the filament for insects to brush against them
- landing platform to ensure contact with anthers and stigma
- mimicry to attract (male) insects

Wind pollinated flower (anemophilus)

- anthers/stigma hang outside the flower to increase chances of pollination
- the style/filament is long to expose stigma/anthers
- stigma is hairy/feathery/branched to increase surface area over which pollen grains land/to trap pollen grains
- pollen grains are smooth/dry/light/small to be easily carried by wind
- large amount of pollen grains to increase chances of pollination
- anthers loosely attached to filaments to enable them to sway to release pollen grains
- pollen grains may have structures which contain air to increase buoyancy
- flowers have long stalks holding them out in the wind

vi) State the ways in which plants prevent self-pollination

- protandry (anthers/stamens mature first)
- protagyny (pistils mature first)
- monoecism (where male and female parts are on same plant but different parts)



- dioecism (where male and female parts are on different plants)
- incompatibility (self sterility)
- heterostyly (styles at different heights)

vii) Give the characteristics that ensure cross pollination takes place in flowering plants

- presence of special structures that attract agents of pollination
- protandry/dichogamy
- protagyny/dichogamy
- monoecism
- self sterility
- heterostyly

viii State the advantages of cross pollination

- hybrid vigour
- less prone to diseases
- promotes genetic variation
- greater evolutionary potential

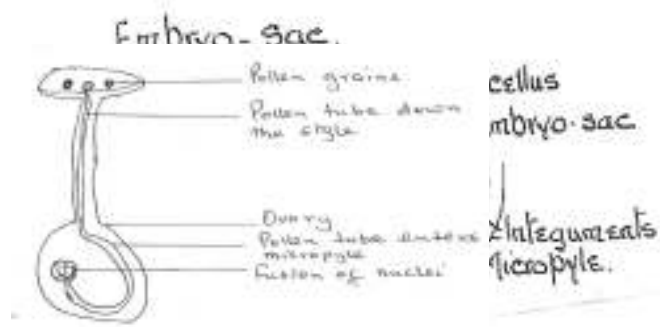
h) i) What is fertilization?

- Fusion of male and female gametes to form a zygote

ii) Describe how fertilization takes place in a flower

- this follows pollination
- pollen grain is deposited on the stigma
- pollen grain sticks to the surface of the stigma
- the surface of the stigma produces a chemical substance which stimulates the pollen grain to produce a pollen tube/to germinate
- the pollen tube grows through the style tissues on which it feeds until it enters the ovary
- the generative nucleus divides into two giving two male nuclei
- embryo sac contains eight nuclei i.e. two synergids, egg cell, two polar nuclei and three antipodal cells
- the pollen tube enters the embryo sac through the micropyle and one of the male nucleus fuses with the egg cell/ovum to form a zygote

- the other male nucleus fuses with the two polar nuclei to form the triploid nuclei/endosperm) food storage used by developing embryo)
- the pollen tube nucleus in the pollen tube disintegrates soon afterwards
- this process is referred to as double fertilization
- zygote grows into an embryo containing plumule, radicle and cotyledons



iii) What is double fertilization?

- there are two male nuclei entering embryo sac
- one fuses with the ovum to form a zygote, while the other fuses with the polar nuclei to form a triploid primary endosperm nucleus
- therefore there are two fusions at fertilization

iv) Name the changes that Occur in a flower after fertilization

- petals, stamen, calyx and style wither
- ovary wall changes into pericarp
- integument changes into seed coat/testa
- zygote changes into embryo (by mitosis)
- primary endosperm nucleus changes into endosperm
- whole ovule changes into seed
- ovary develops and grows into fruit (under the influence of gibberellic hormone)

b) i) Distinguish between a fruit and a seed

- a fruit is a fertilized ovary and has two scars
- a seed is a fertilized ovule and has one scar

ii) How is a seed formed?

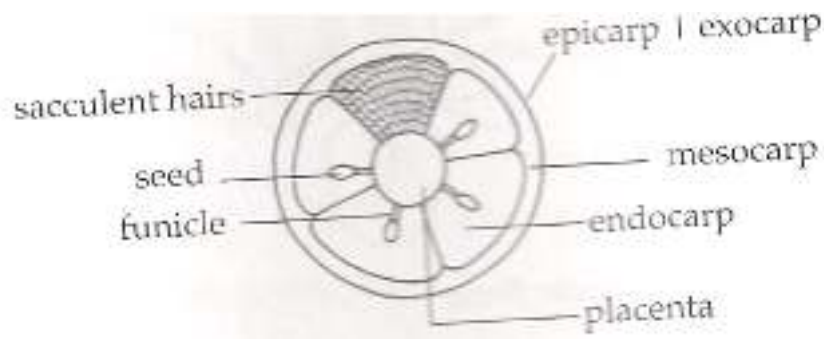
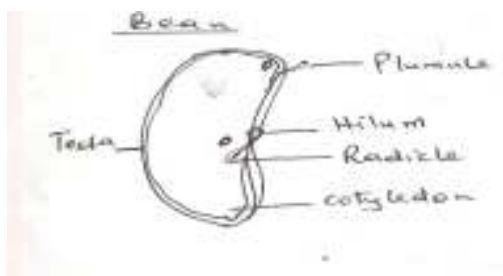


- after fertilization, zygote grows into an embryo, primary endosperm nucleus developed into endosperm, integuments harden to form testa, hence the whole ovule becomes the seed
- the seed loses water to become drier
- the seed has plumule, radicle, seed leaves called cotyledons, a microphyll and a scar

- store food for the germinating seed i.e. for plumule and radicle
- when plumule and radicle grow, they use food stored in the cotyledon
- in some seeds food is stored in the endosperm

v) Draw a labeled diagram of a fruit

iii) Draw a labeled diagram of a seed



iv) Describe the main parts of a seed

Testa

- also called seed coat
- a tough outer covering which protects the seed from insects, bacteria etc
- segment is the membrane inside the testa

Hilum

- a scar
- spot where the seed was attached to the fruit or pod

Micropyle

- small hole through which water and air enter the seed

Radicle

- embryonic root
- grows into the shoot system

Cotyledons

- embryonic leaves

vi) How is a fruit formed?

- one of the organs that remains on the plant after pollination and fertilization is the ovary
- within the ovary, the developing embryo produces special chemical substances that stimulate the young ovary
- these substances also signal the start of the formation of the fruit, which is a mature ovary
- the fruit may contain one or more seeds
- during fruit formation the ovary increases in size while ripening or maturing
- a true fruit is formed from the ovary of a flower after fertilization
- it has two scars (style scar and stalk scar) and contains seeds
- some seeds are not formed from the ovary of a flower



- some other parts of a flower develop to form a fruit
- such fruits are called false fruits

vii) Explain the importance of fruits in the survival of plants

- protect the seed against dessication, predators and adverse conditions
- aid in seed dispersal by attracting agents of dispersal
- stores food for the plant

vii. Distinguish between parthenogenesis and parthenocarpy

- parthenogenesis is development of new animals from unfertilized eggs
- parthenocarpy is development of a fruit without fertilization

iv) State the differences between a seed and fruit

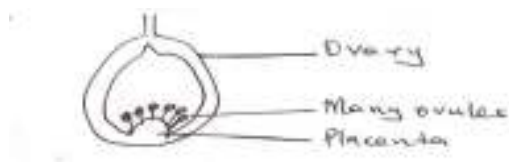
Seed	Fruit
• fertilized ovule • attached to placenta through funicle • one scar called hilum • has seed coat/testa • seed wall undifferentiated	• fertilized ovary • attached to branch through a stalk • two scars (style scar and stalk scar) • has fruit wall/pericarp • fruit wall is differentiated

j. i) What is placentation?

- arrangement of ovules within the plant ovary

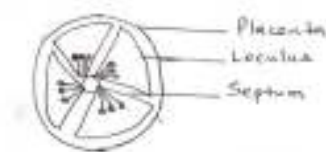
ii) Explain the following types of placentation

Marginal



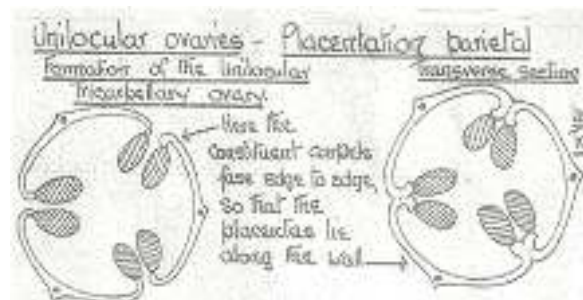
- placenta appears as one ridge on ovary wall
- ovules are attached to placenta in rows e.g. peas in a pod

Basal



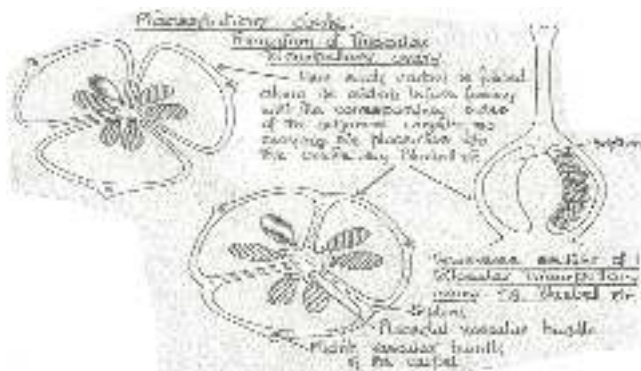
- placenta formed at the base of the ovary with numerous ovules attached to it

Parietal



- edges of carpels fuse together
- dividing walls disappear, leaving one loculus
- have numerous seeds e.g. passion fruit
- placenta of each carpel appears as ridges on ovary wall

Axile



- formed from a single flower or one ovary e.g. mango

- consists a group of ovaries that appear on a common receptacle e.g. strawberry

- edges of carpels fuse together to form a single central placenta
- numerous ovules arranged on placenta
- ovary divided into a number of loculi by walls of the carpel e.g.

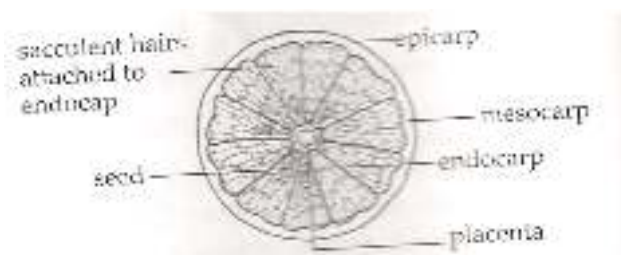
[illegible]

- edges of carpels fuse together
- dividing wall disappears leaving one loculus

- formed from several flowers whose ovaries fuse together after fertilization
- form a bunch e.g. pineapple, figs
- are always false fruits

- also called fleshy fruits
- all or part of pericarp (fruit wall) becomes juicy

-Berry

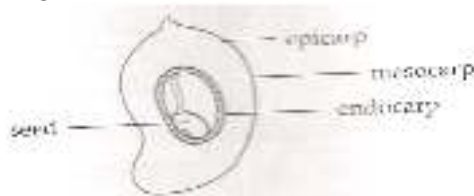




- has many seeds
- whole pericarp is succulent e.g. orange, tomato, pawpaw

Drupe

- only one seed
- pericarp divided into three layers i.e. epicarp, mesocarp(juicy) and endocarp(hard)
- e.g. mango and coconut



Pome

- juicy part is swollen receptacle
- is usually a false fruit
- example is a pear

iv) What are dry fruits?

- have a pericarp that is dry, hard and woody
- either dehiscent or indehiscent
- called dry because they are not succulent

v) What are dehiscent fruits?

- split open when ripe to release seeds
- contain many seeds

vi) Give types of dehiscent fruits

Legumes

- split along two edges

- are usually pods e.g. beans, peas, crotonaria

Follicle

- split on one side only e.g. Sodom apple

Capsule

- has several lines of weakness/sutures
- open in many places e.g. castor oil, cotton

vii) What are indehiscent fruits?

- non-splitting fruits
- usually one seeded only

ii) Give main types of indehiscent fruits

Nut

- pericarp woody, hard and thick e.g. cashew

Achene

- has thin, tough pericarp e.g. sunflower

d) i) What is seed and fruit dispersal?

- spreading of seeds and fruits away from parents so as to settle where conditions are suitable for their germination

ii) Why is dispersal of seeds and fruits necessary?

- prevent overcrowding
- reduces competition for space, nutrients and light
- colonization of new areas is made possible
- to increase chances of survival
- to prevent inbreeding
- to avoid extinction due to over competition for the necessities

iii) Explain how seeds and fruits are adapted to various methods of dispersal

Adaptations for wind dispersal

- they have wings, feathers or hair-like structures to increase surface area for wind to carry them easily/buoyancy
- seeds/fruits are loosely attached on the stalks so that they can easily be released and carried away by wing
- seeds/fruits are generally light and small sized to be easily carried by wind



- some seeds/fruits have parachute-like structures to be easily carried by wind
- some have sensor mechanism where seeds and fruits are borne on long stalks that are loosely attached which allows swaying so that movements of capsule by wind releases the seeds

Water dispersal seeds

- seed/ mesocarp has air spaces thus light/buoyant to float hence carried by water
- they have waterproof cover and tough pericarp protects seeds from getting soaked
- fibrous and spongy mesocarp to easily float

Animal dispersal seeds

- presence of hooks for attachment to animals thus carried to other parts
- fruits are brightly coloured, succulent and scented to attract animals
- seed coats are hard and resistant to digestive enzymes hence seeds are dropped away from mother plant
- large in size or borne on clusters to be easily seen

Self dispersal/explosive

- self opening seeds
- they have lines of weakness called sutures for violent opening thus scattering seeds away from parent plant

5. a) i) Distinguish between external and internal fertilization in animals

- in external fertilization fusion of the male and female gametes takes place outside the body of the female e.g. amphibians and fish
- in internal fertilization union of gametes occurs inside the body of the female

ii) State the advantages and disadvantages of external fertilization

Advantages

- large numbers produced therefore many offspring per breeding season
- female does not suffer gestation stress
- mother does not need to care for the young except in a few species
- the surviving individuals are highly selected for better survival

Disadvantages

- many predators surround the eggs before and after fertilization
- fewer chances of fertilization/a lot of gametes wasted
- embryo development at mercy of environment
- large numbers of female gametes are required therefore female gets much exhausted

iii) State the advantages and disadvantages of internal fertilization

Disadvantages

- number of gametes fewer hence less number of offspring
- less adapted for sudden change of environment after birth
- in mammals females suffer gestation stress

Advantages

- more chances of fertilization
- fewer predators of oval/fertilized egg protected in females body
- stable internal environment
- fewer gametes required

iii) Give a reason why it is necessary for frogs to lay many eggs

- to increase chances of survival/fertilization

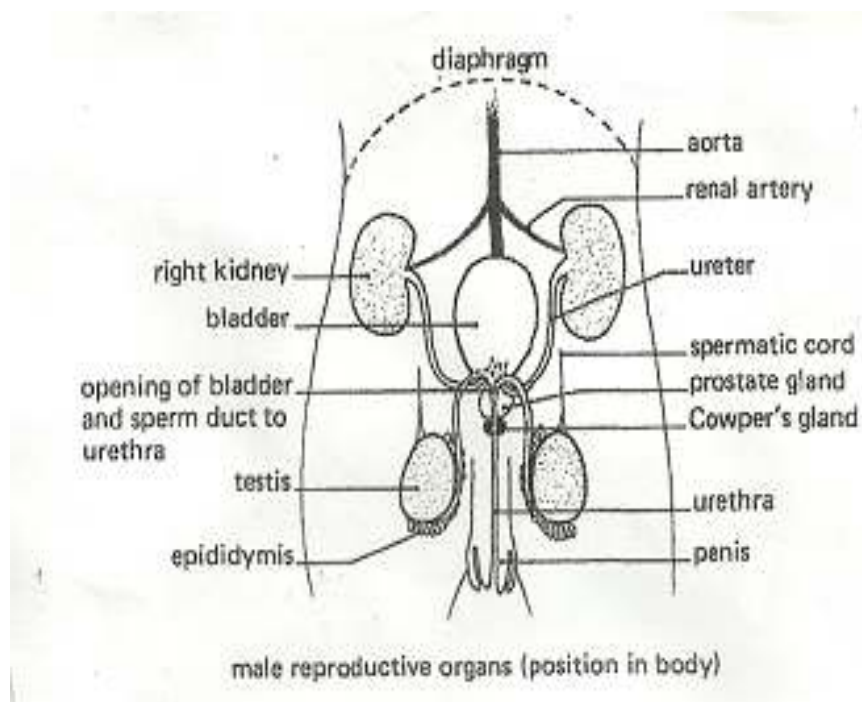
iv) Compare external and internal fertilization

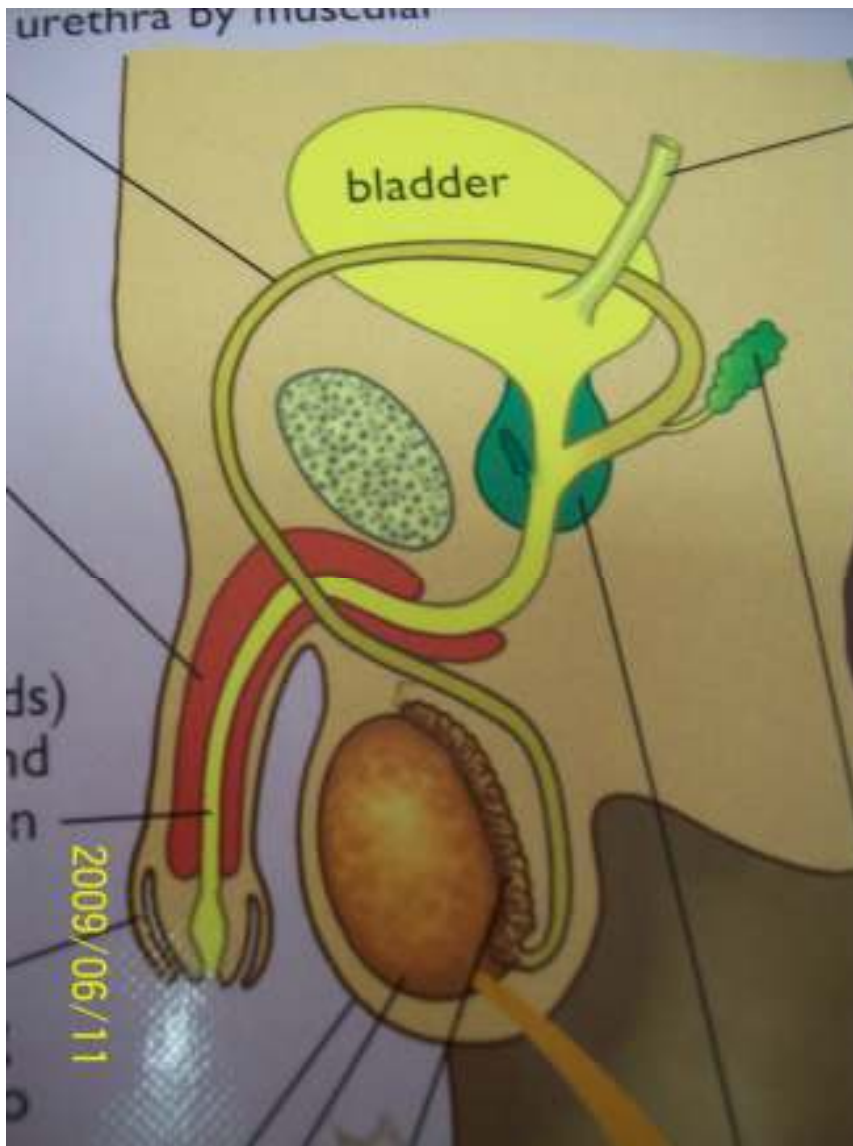
External	Internal
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• occurs in water outside the bodies of animals • many eggs are laid • usually less contact between male and female • both fertilized eggs exposed to danger	• occurs inside the body of the female animal • fewer eggs released from ovary • very close contact in form of copulation between male and female • Fertilized eggs are enclosed hence highly protected inside females' body.
--	---

b) i) Draw and label the human male reproductive system





ii) Describe how the mammalian male reproductive system is adapted to perform its functions

Penis

- is highly vascularised/spongy
- has sensitive glands
- becomes erect to allow entry into the vagina

Scrotum

- contains the testes outside the body on whose walls the process of spermatogenesis takes place
- the process is favored by lower temperature
- it contains sertoli cells which nourish sperms until they are mature

Epididymis

- long and coiled for the purpose of sperm storage

Vas deferens



- muscular
- upon contraction pushes sperms out and allows ejaculation

Gametes

- produced in large numbers to increase chances of fertilization
- the sperms have a tail for swimming/large number of mitochondria to provide energy/allow swimming to reach the egg

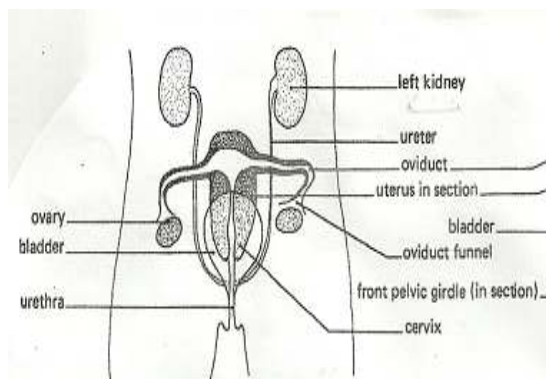
Accessory glands

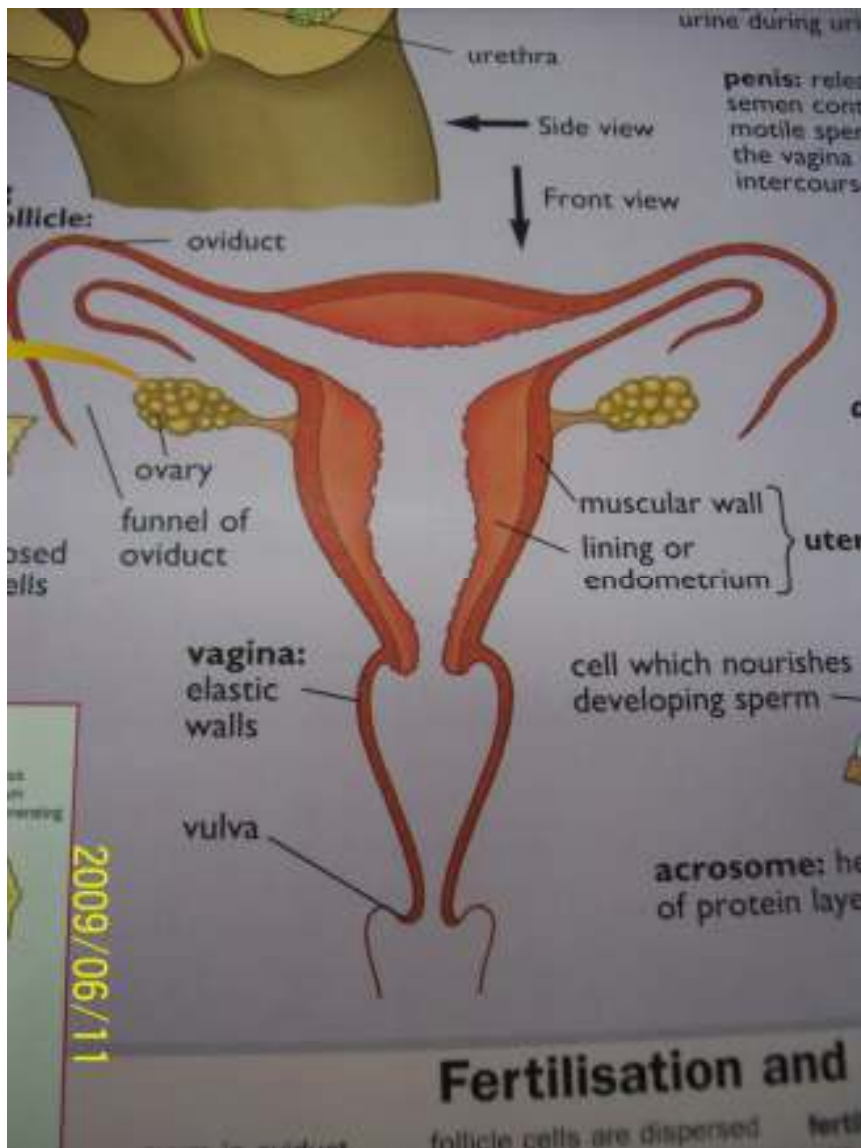
- are seminal vesicle, Cowper's gland and prostate gland
- they produce seminal fluid to provide a medium/ nutrients for sperms to swim

iii) How is the sperm adapted to perform its function?

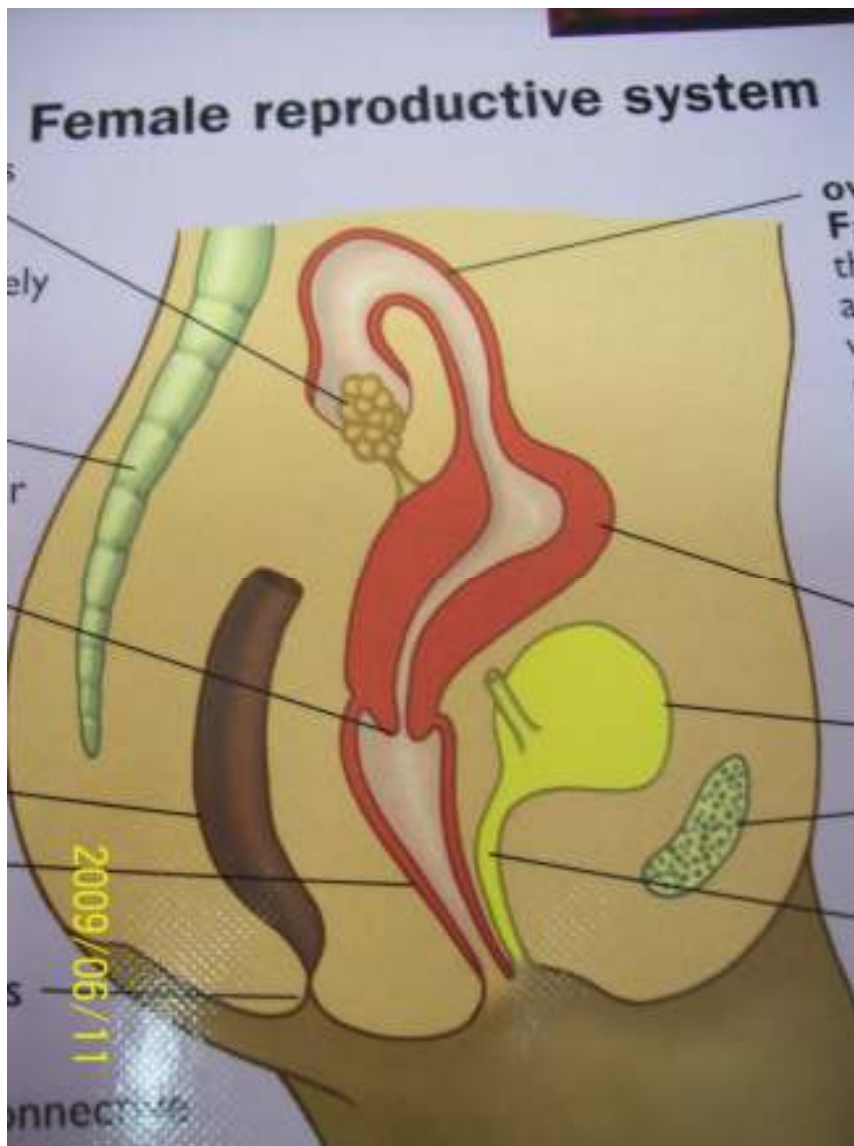
- acrosomes contain enzymes to digest egg membrane
- nucleus contains genetic material
- mitochondria produce energy to move the tail back and forth
- the lashing movement of the tail enables the sperm to move/propulsion in fluid medium towards the egg
- it is streamlined for faster/easier movement/swimming to meet the egg

c) i) Draw and label the human female reproductive system









ii) Describe how the various structures of the human female reproductive system are adapted to their function

Ovaries

- have several graafian follicles that develop and burst open to release/produce mature ova
- secretes sex hormones (oestrogen) which initiate/control development of secondary sexual characteristics
- produce hormones oestrogen and progesterone which prepare the uterus for implantation and subsequent nourishment of the embryo

Oviducts (fallopian tube)

- are thin narrow and tubular to increase flowing speed of semen containing sperms
- are funnel shaped on the end next to ovary which enables them to receive the ovum
- their lining contains cilia which propel the ovum towards the uterus
- has peristaltic muscles that enable movement of zygote/ovum to the uterus for implantation
- is fairly long to increase surface area for fertilization

Uterus

- is muscular for protection of developing embryo
- has elastic wall that allows growth and development of foetus/embryo
- has a highly vascularised endometrium that provides nutrients/gaseous exchange to developing embryo

Cervix



- has valves that close the lower end of the uterus to ensure continued pregnancy during gestation period
- is capable of dilating
- has narrow entrance/neck-like entrance to uterus that enables quick swimming of sperms to uterus
- has suction mechanism that draws up/pulls sperms into uterus
- has a “W” shape that fits well with the glands of the penis to ensure sperms are deposited at the right point

Vagina

- is elastic and muscular to enable good accommodation or penetration of the penis thus proper deposition of sperms and for easy parturition
- allows menstrual flow
- has sensitive labial walls which secrete/produce lubricating substances that ensure/enable/facilitate good coition
- capable of considerable enlargement, due to elastic muscles, to accommodate baby during parturition

Clitoris

- has sensitive cells for orgasm

iii) Explain how the ovum is adapted to its function

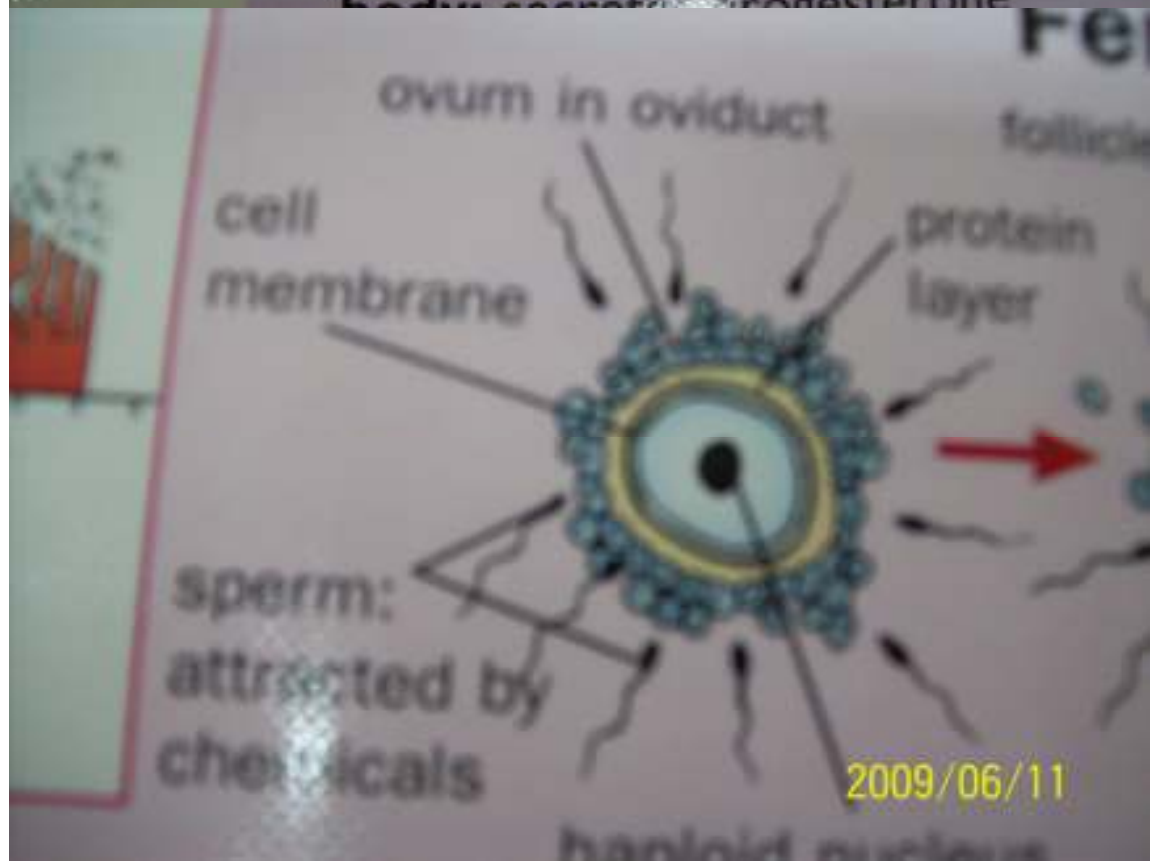
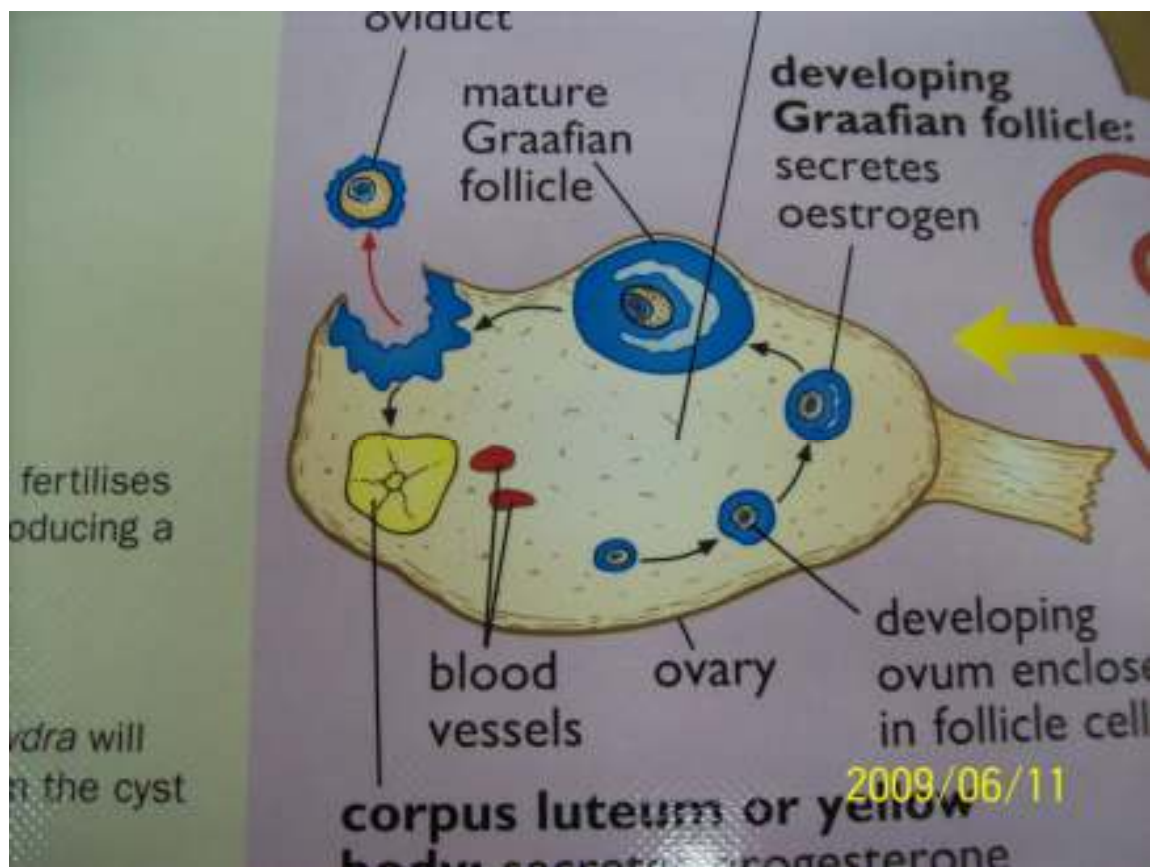
- nucleus contains genetic material
- ventelline membrane encloses plasma membrane which encloses yolky cytoplasm
- yolky cytoplasm provides nourishment
- jelly coat protects ovum against dehydration

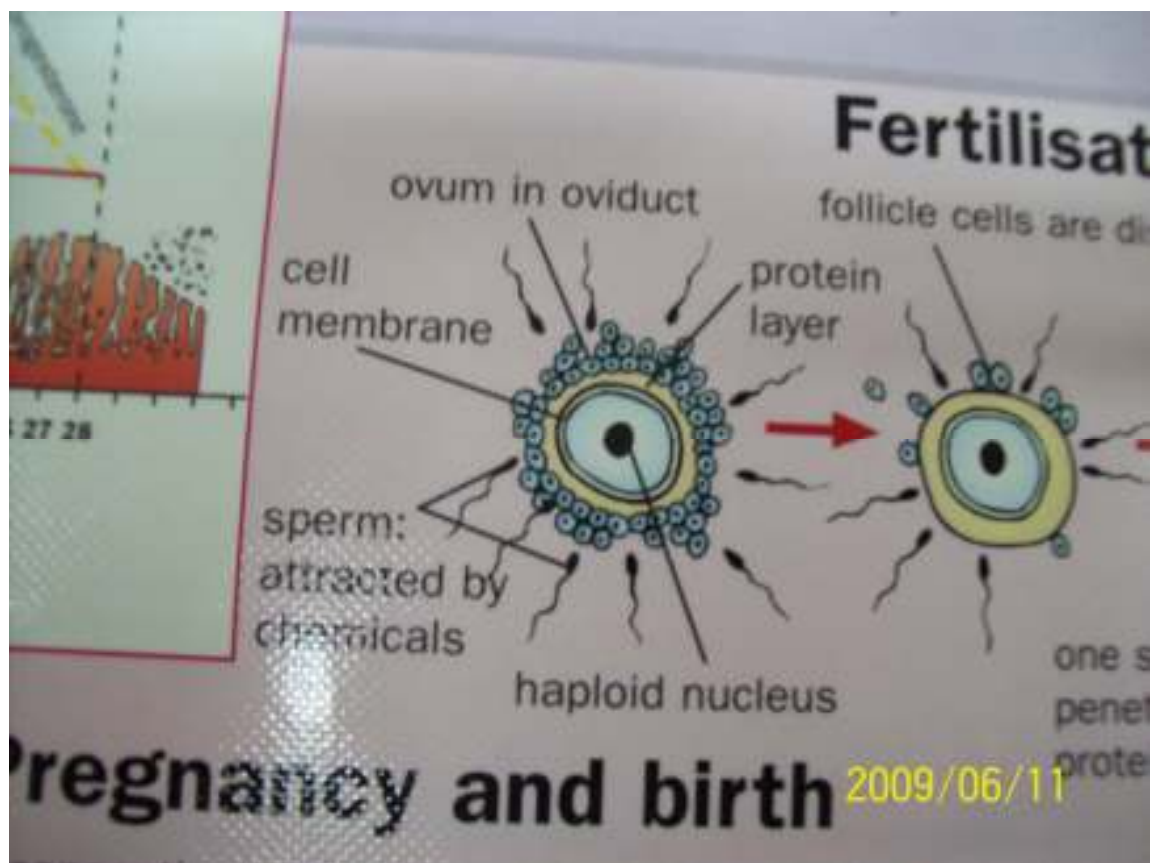
iv) Explain the differences between sperm and ovum

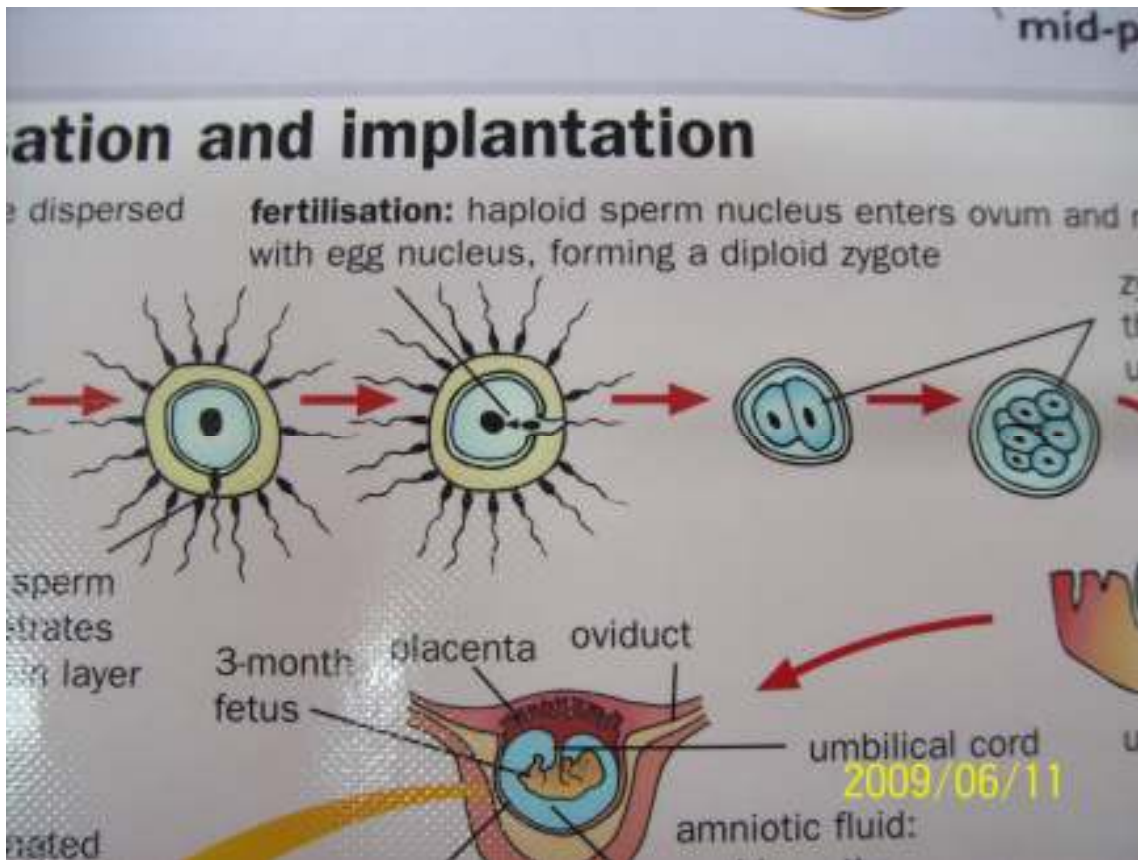
Sperm	Ovum
• long with a tail and head • small • locomotory • stores little food • has acrosome (tip with lytic enzymes) • nucleus prominent but cytoplasm negligible	• spherical • large • stationary • a lot of food stored in yolky cytoplasm • lacks acrosome • a lot of cytoplasm

d) i) Explain the process of fertilization

- a process whereby the egg and sperm are brought together and fuse to form a zygote
- occurs in the fallopian tube after copulation
- sperm head penetrates the outer coat of the ovum while the tail remains outside
- penetration is due to reaction of acrosome
- acrosome digests the vitelline membrane
- thereafter a zygote is formed
- zygote which is diploid undergoes rapid cell division to form a mass of cells called blastocyst
- after fertilization a membrane forms around the ovum to prevent further entry of sperms
- blastocyst eventually develops into an embryo

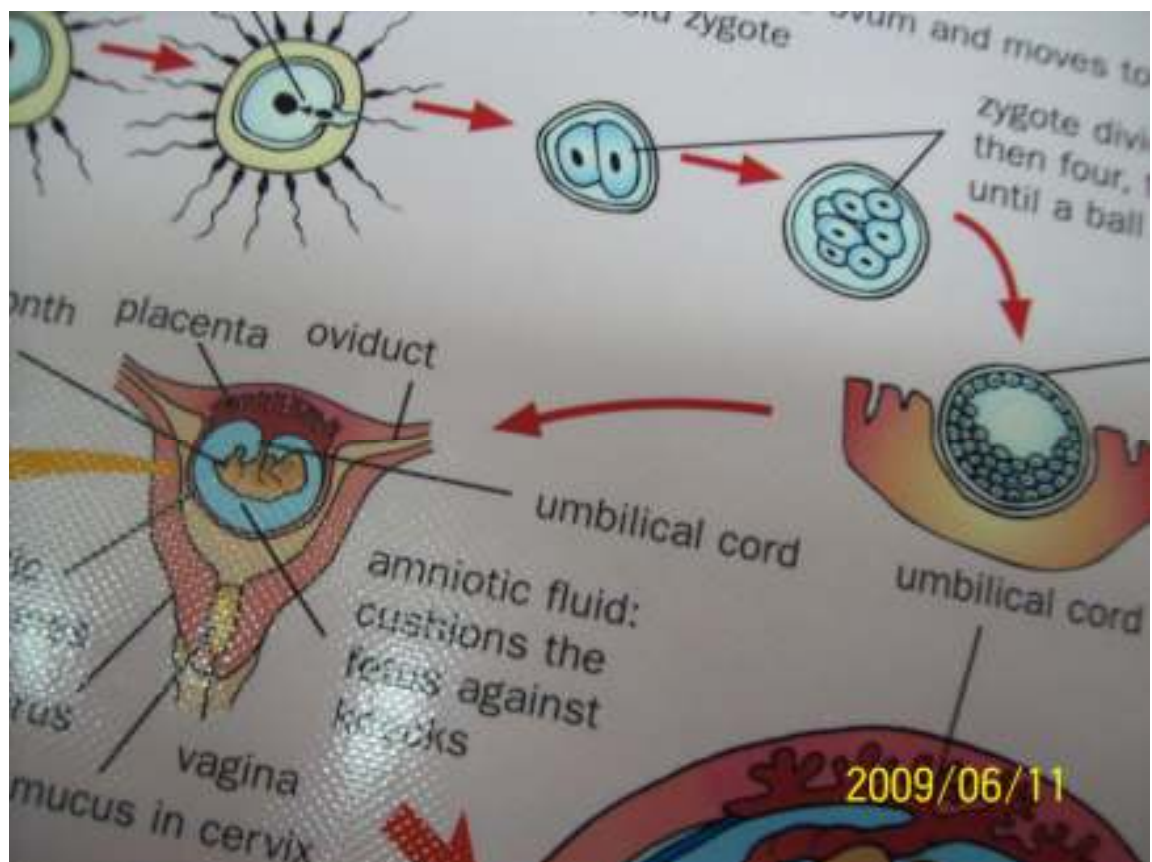


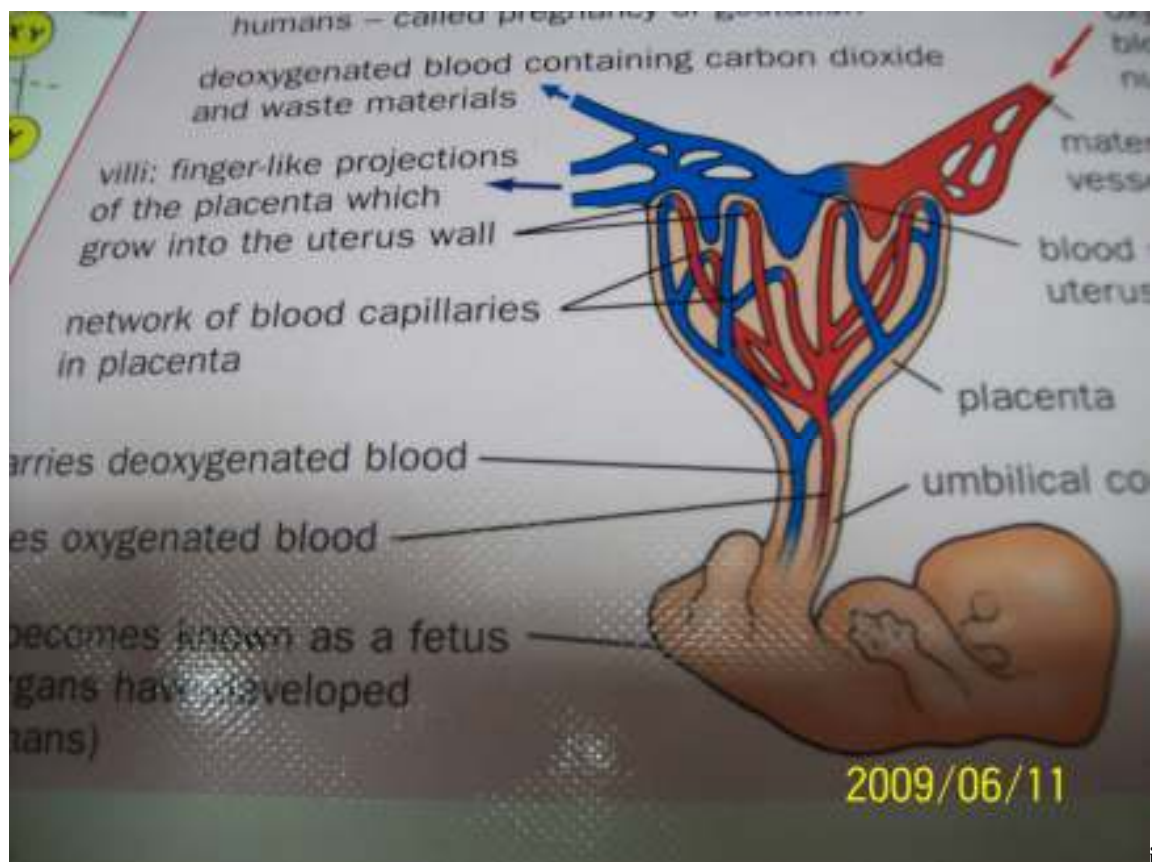


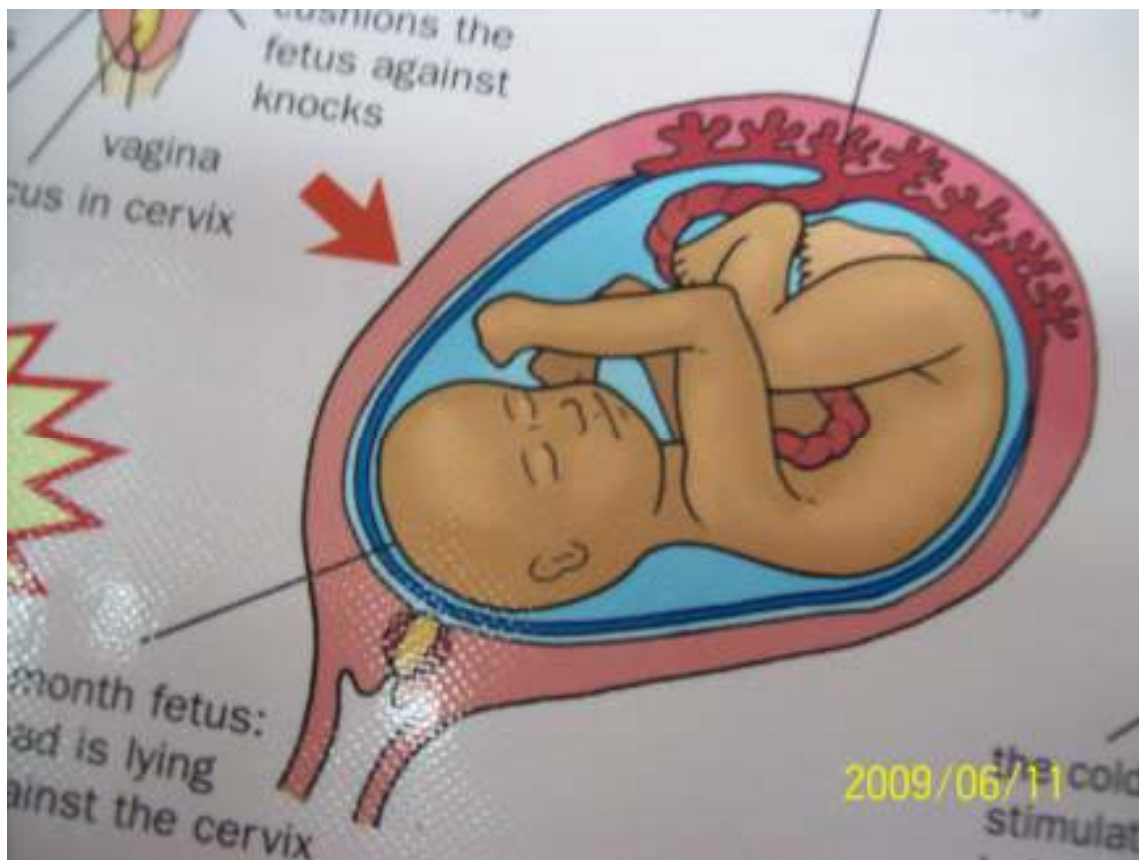


i) Explain the process of implantation

- this is the embedding and attaching of the embryo in the uterine wall/endometrium
- implantation marks the beginning of pregnancy
- sometimes implantation occurs in the oviduct wall which is abnormal and results in ectopic pregnancy which is fatal
- the outer wall of the blastocyst develops fingerlike projections which project into the uterine wall for attachment
- the projections are called villi
- the villi and endometrium develop into an organ that is called the placenta
- the embryo is attached to the placenta through a cord called the umbilical cord







State the functions of umbilical cord

- it contains blood vessels (umbilical artery, iliac arteries and umbilical veins)
- it joins the placenta to the embryo
- passage for nutrients from the mother
- passage of excretory substances from foetus to mother for final discharge
- gaseous exchange
- passage of antibodies from mother to foetus, for protection of foetus against diseases

iv) State the role of placenta

- exchange of gases between mother and foetus
- exchange of nutrients and nitrogenous wastes
- anchorage/attachment of foetus
- produces hormones (oestrogen and progesterone)

e) i) What is gestation period?

- time taken from fertilization to birth/pregnancy

ii) Explain the functions of the membranes associated with placenta

Chorion

- surrounds the embryo
- has fingerlike projections that attach embryo to the uterus

Amnion

- contains amniotic fluid
- fluid surrounds embryo
- protects embryo from mechanical injury by acting as shock absorbers
- fluid also protects embryo from dehydration
- distributes pressure equally over embryo

Yolk sac

- surround the yolk
- produces blood cells for embryo until its own liver is able to perform the task

Allantois

- present only for a short time
- removes and store waste material
- it eventually becomes the umbilical cord

iii) Explain the events that take place to facilitate parturition



- near birth the placenta produces less progesterone
- oxytocin hormone is produced by posterior lobe of pituitary gland
- because progesterone level has decreased the uterus becomes sensitive to oxytocin
- oxytocin causes the contraction of the uterus (myometrium)
- these contractions are called labour pains
- just before parturition the head turns downwards
- the contractions eventually push the baby through the vagina
- amnion breaks and amniotic fluid is released
- oxytocin dilates the cervix
- foetus is expelled through cervix with head coming out first
- finally the whole infant comes out
- the umbilical cord is cut and the placenta is expelled as afterbirth

iv) State the reasons why later in pregnancy the ovary will be removed without disturbing the pregnancy

- corpus luteum in the ovary secretes progesterone which maintains pregnancy and development of foetus after conception
- after four months pregnancy is maintained by progesterone from the placenta

f) i) What are secondary sexual characteristics

- Characteristics (physiological and anatomical) that start developing at puberty due to the influence of male and female hormones

ii) State the main secondary changes in

Boys

- deepening of the voice
- growth of hair on face, pubic part, chest, legs
- penis and testes become bigger
- muscular development
- sperm production begins at puberty and may continue throughout life

Girls

- growth of hair on pubic part and armpits
- widening/enlargement of hips
- development of breasts
- menstrual cycle starts as ovaries mature

- body acquires extra fat

iii) Describe the role of hormones in secondary sexual characteristics in

Boys

Follicle stimulation hormone (FSH)

- from pituitary
- stimulates production of androgens (male hormones) mainly testosterone by testis

Testosterone

- secondary sexual characteristics

Girls

FHS

- from pituitary
- development of follicles
- stimulates oestrogen production by ovary

LH

- from pituitary
- ovulation
- stimulates release of progesterone by ovary

Oestrogen

- stimulates release of LH
- secondary sexual characteristics

Progesterone

- also from placenta
- sustains pregnancy as it inhibits prolactin and oxytocin during pregnancy

Prolactin

- milk formation

Oxytocin

- parturition
- milk ejection

g) i) What is menstruation?

- vaginal discharge due to disintegration of endometrium

ii) Describe the role of hormones in the human menstrual cycle



- it is controlled by sex hormones which are responsible for the onset of secondary sexual characteristics and also control of the menstrual cycle
- the onset is signaled by discharge of blood/menses 14 days following the start of menstruation
- anterior lobe of pituitary gland secretes follicle stimulating hormone (FSH)
- Follicle stimulating hormone causes graafian follicle to develop in the ovary. It also stimulates tissues of the ovary/wall (theca) to secrete oestrogen
- oestrogen causes repair/healing of uterine wall
- oestrogen stimulates anterior lobe of pituitary to produce luteinising hormone (LH)
- LH causes ovulation. It also causes graafian follicle to change into corpus luteum. LH stimulates corpus luteum to secrete progesterone
- Progesterone causes proliferation/thickening of uterine wall
- Oestrogen/progesterone inhibits the production of FSH by anterior lobe of pituitary, thus no more follicles develop and oestrogen production reduces
- In the next two weeks progesterone level rises and inhibits production of LH from anterior lobe of pituitary
- The corpus luteum stops secreting progesterone and menstruation occurs when the level of progesterone drops
- Anterior lobe of pituitary starts secreting FSH again.

iii) What is menopause?

- end of ovulation in women
- occurs after age of 45 years
- does not occur in males

h) Explain the symptoms, methods of transmitting and prevention (control) of the following sexually transmitted diseases

i) Gonorrhea

- caused by a bacterium called Neisseria gonorrhea
- transmitted through sexual intercourse,
- infects urethra and vaginal tract (epithelia)
- Symptoms include pain, discharge of mucus and bad smell (females)

- Effects include sterility, heart diseases, blindness of foetus and arthritis
- Treatment by antibiotics
- Control and prevention by proper sexual conduct

ii) Herpes

- notably Herpes simplex and H. genitalis
- caused by virus which attacks genitalia
- symptoms are painful sores in genitalia, skin lesions
- transmitted in saliva, sexual intercourse and injection by drug addicts
- no treatment

iii) Syphilis

- caused by bacterium called Treponema palladium
- symptoms are painless wounds in genitalia
- attack genitalia, nervous system, lips
- treated by antibiotics

iv) Trichomoniasis

- caused by plasmodium called Trichomonas
- attacks reproductive tract
- symptoms are itching and discharge of pus from the genitals
- treated by antibiotics

v) Hepatitis

- Viral disease

Affects the liver

Transmitted through sexual intercourse

No known treatment



vi) Candidiasis

- caused by fungus called *Candida albicans*
- transmitted through sexual intercourse
- symptoms include itching urethra, and vaginal discharge (odourless)
- controlled by personal hygiene, early treatment and responsible sexual behavior

vii) HIV/AIDS

- caused by HIV virus
- transmitted by sexual intercourse, blood transfusion, sharing piercing instruments from infected mother to foetus, infant and baby
- symptoms include fever, swollen lymph nodes, night sweating, cough, weight loss, fatigue, loss of appetite, diarrhea, headache, opportunistic infections and tumors
- Control by responsible sexual behaviour, education, screening blood for transfusion and using sterile piercing instruments.

6. a) Define the terms

i) Growth

- an irreversible change in size of a cell, organ or whole organism
- growth is due to synthesis of protoplasm or extracellular substances

ii) Development

- refers to a series of changes which an organism goes through in its life cycle
- during development both qualitative and quantitative changes take place (involves differentiation)

iii) Differentiation

- refers to changes in which the cells of the body undergo and become specialised to perform specific functions

b) i) Differentiate growth in plants and animals

Plants	Animals
- confined to shoots/root tips (apical) - have definite	- occurs all over the body (intercalary) - different parts

growth regions (meristems) - often indefinite/continuous - growth results in branching - affected by light, auxins, hormones and temperature	grow at different rates (allometric) - maximum growth on maturity (definite) - no branching - affected by nutrients
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ii) List the processes involved in growth

- assimilation
- cell enlargement
- cell division (by mitosis)

iv) List the parameters used to measure growth

- height/length
- dry weight
- number of individuals
- volume
- leaf area of plant

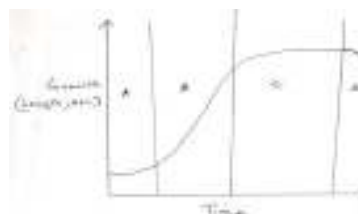
iv) Name the patterns of growth in organisms

- allometric and isometric
- limited and unlimited
- discontinuous growth

c) i) Name the different types of growth curves

- sigmoid curve (normal growth curve)
- intermittent growth curve

ii) Draw a sigmoid growth curve and explain its different phases/stages



A-lag phase

- Slow growth rate at first

Organism adapting to the environment



B-exponential phase

- organisms already adapted
- first growth due to birth rate that is higher than death rate

C- Stationary phase (plateau)

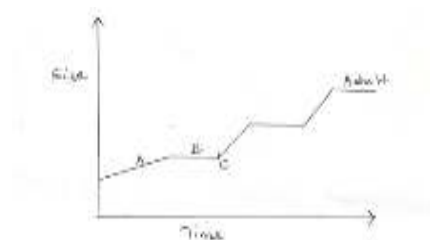
- Birth rate equals death rate (equilibrium)

Lack of nutrients, accumulation of toxic waste products

D-phase of decline

- due to depletion of nutrients, accumulation of toxic wastes, lack of space
- some individuals old hence not reproducing
- death rate higher than birth rate

iii) Draw an intermittent curve and explain the various stages



A-growth

B-no growth

C- moulting/ecdysis

- seen in arthropods
- growth in arthropods is intermittent(takes place during some time only because their hard cuticles (exoskeleton) does not expand to cause growth
- the cuticle must be shed off first to allow further growth
- the shedding is called ecdysis or moulting
- when moulting has taken place animal grows but growth stops when the exoskeleton hardens again

d) i) What is seed dormancy?

- A state where a viable seed is incapable of germinating when all conditions are favourable.

ii) State the biological importance of seed dormancy

- gives embryo time to reach maturity
- gives time for dispersal
- allows plant to survive adverse conditions

iii) State the factors which cause seed dormancy

Internal factors

- presence of abscisic acid/ABA/ presence of germination inhibitors
- embryo not fully developed
- absence of hormones/enzymes/inactivity of hormones/enzymes/gibberellins/cytokinins
- impermeability of seed coat

External factors

- unsuitable temperature
- absence of light
- lack of oxygen
- lack of oxygen
- lack of water

iv) Give the conditions necessary to break seed dormancy

- scarification/scratching to make seed coat impermeable
- vernalisation/cold treatment in some seeds like wheat
- burning/nicking/expose to heat e.g. wattle seeds
- destruction of germination inhibitors

e) i) What is seed germination?

- process by which a seed develops in a seedling

ii) What is viability

- ability of a seed to germinate



iii) Discuss the various conditions necessary for the germination of seeds

Water

- medium for enzymatic activity
- hydrolysis of food into simpler substances
- medium of transport
- softens the seed
- acts as a solvent

Air

- in form of oxygen
- oxygen is used for respiration/oxidation of food to release energy

Suitable (optimum) temperature

- activates enzymes involved in mobilization of food reserves

Enzymes

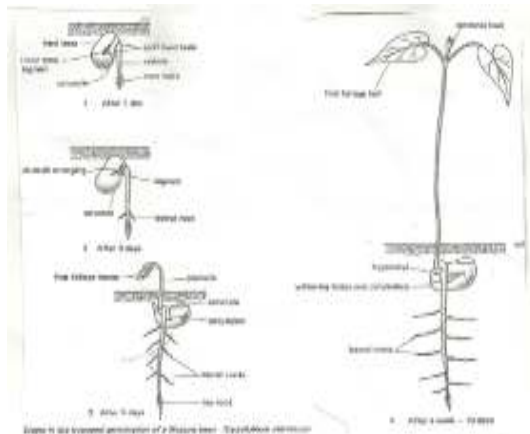
- breakdown and subsequent oxidation of food
- conservation of hydrolyzed food products into new plant tissues

Viability

- only viable seed are able to germinate and grow

iv) Name and describe the types of germination

Epigeal



cotyledons are brought above the ground level during germination due to elongation of hypocotyls

Hypogeal

- the cotyledons remain below the surface during germination due to elongation of epicotyl e.g. maize

v) Name the part of the bean seed that elongates to bring about epigeal germination

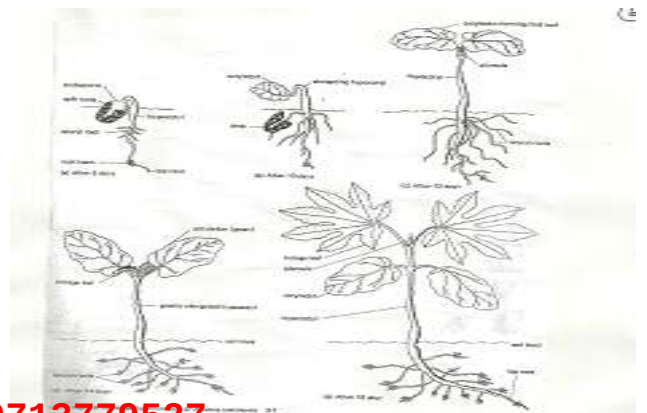
hypocotyl

vi) Account for the loss in dry weight of cotyledons in a germinating bean seed

- food stored is mobilized/used up for respiration and growth

vii) Describe the physiological changes that occur in a seed during germination

- in presence of oxygen, optimum temperature and water, food reserves in the seed are hydrolysed or broken down into soluble diffusible form by enzymes
- soluble food diffuses to the growing embryo
- oils and carbohydrates provide energy
- simple sugars converted to cellulose to form cell wall
- amino acids make protoplasm
- seed develops plumule and radicle hence





germinates

viii) Explain the biological significance of cotyledons being brought above the ground in epigeal germination

- cotyledons have inadequate food
- they are brought above the ground to acquire chloroplasts to carry out photosynthesis before the formation of foliage leaves to supplement food supply required for growth during germination

f) i) Distinguish between primary and secondary growth

Primary growth

- occurs at the apical (shoot and tip) apices regions where meristematic cells occur
- causes plant elongation since cells divide by mitosis

Secondary growth

-occurs at the cambium meristems

Increases width (girth) of the stem

ii) What are meristems

- dividing cells
- meristem means they are dividing

iii) State the characteristics of meristematic cells

- dense cytoplasm
- thin cell walls
- absence of vacuoles/cell sap

iv) State the location and function of the following meristematic tissues

Apical meristem

- located at tips of roots and shoots
- increase length of stem and roots/primary growth

Intercalary meristem

- found at bases of internodes
- responsible for elongation of internodes and increase in leaf sheath in grasses

Lateral meristems

- found near the periphery of stem and root
- responsible for secondary growth/growth in girth of stem and root/lateral growth
- called cambium and constitute vascular and cork cambium

v) Describe primary growth

- occurs at tips of shoots and roots in the meristematic tissues of apical meristem
- at the apex there is a zone of cell division/mitosis
- cells elongate at elongation zone
- the elongated cells differentiate at the region of differentiation resulting in increase in size
- in the stems meristems give rise to leaf primordia which envelop the apex to form a bud
- the bud protects the delicate inner cells
- in roots the meristem is protected by root cap
- after cells differentiate they form permanent tissues

vi) Describe secondary growth in plants

- also called secondary thickening
- only occurs in dicotyledonous plants that have cambium
- monocotyledonous plants do not undergo secondary growth because they lack intervacular cambium
- cambium cells divide to produce more cells on either side of the cambium
- cells produced to the inside become secondary cambium
- cells produced to the outside become secondary phloem
- division of cambium cells occurs yearly producing new rings of secondary phloem and secondary xylem each year
- intervacular cambium(cambium between vascular bundles) divide to form secondary



parenchyma, thereby increasing growth of medullary rays

- much more xylem is formed than phloem, thus pushing phloem and cambium ring outward
- the rate of secondary growth is depended on seasons(rains) resulting in annual rings
- cork cambium is located beneath epidermis and is responsible for secondary thickening of the bark of perennial plants
- cork cambium divides to form new cork(bark) tissues to accommodate increased growth on outside and secondary cortex on the inside
- Cork cells (cells of the bark) are loosely parked at some points to form lenticels for gaseous exchange.

vii) State the significance of secondary growth

- increase girth or circumference of trees
- annual rings which show seasonal growth can be used to tell the age of trees

- use the same leaf throughout
- trace the outline on a graph paper and work out the area
- record
- repeat at regular intervals until regular area
- average rate of growth equals to total increase in area divided by the period of time taken to achieve full area

Average rate of growth = $\frac{\text{total increase in area}}{\text{period of time taken to achieve final area}}$

period of time taken to achieve final area

i

i) Describe how the growth of a root can be determined

Materials

- fine thread, marking ink, germinating bean seedlings, blotting paper, ruler marked in millimeters, pins, cork, a boiling tube and moist cotton wool

Procedure

- dry seedlings using blotting paper
- place inside against the ruler marked in mm
- dip the fine thread in waterproof ink
- mark the radicle at equal intervals
- pin the seedling to the cork
- suspend the seedling into the boiling tube containing moist cotton wool
- allow the seedling to grow for two days/some time observe the intervals with the marks
- record your observations the widest intervals are found in the region just behind the tip indicating/showing region of greatest growth

iii) A boy hammered a nail in the bark of a tree at a height of 1.5metres above the ground.

Four years later, the nail was found at the same height although the tree had grown 3 meters taller. Explain the above observation

g) i) Describe one method which can be used to measure the average growth rate of a single leaf of a plant

EITHER

- chose/identify a young leaf(just unfolded)
- use the same leaf throughout
- measure (total) length of (whole) leaf
- record
- repeat at regular intervals until no more change occurs/constant length
- average rate of growth is equal to total increase in length divided by the period taken to achieve full length

Average rate of growth = $\frac{\text{total increase in length}}{\text{period taken to achieve full length}}$

OR

- choose/identify a young leaf(just unfolded)



The nail was hammered at a point where vertical growth had stopped/further growth was confined to increase in width/diameter.

Vertical growth is confined to tips/apex/vertical apical meristem

h) i) Describe the role of hormones in growth and development of plants

indole acetic acid/IAA/ auxins

Cell division/increase in cell division

Tropic responses

Cell elongation/increases in cell elongation

Development of abscission layer

Growth of ovaries into fruits/parthenocarpy/initiates flowers

Inhibits growth of lateral buds/produces apical dominance

Stimulates adventitious/lateral roots

Gibberellins (Gibberellic acid/GA3)

Promote cell elongation/rapid cell division/increase in length of the internodes

Promote fruit formation without fertilization/parthenocarpy

Reduces root growth

Breaks seed dormancy/promotes germination

Cytokinnins (Kinnins/Kinnetin/Zeatin)

- breaks dormancy
- promotes flowering
- promotes cell division
- stabilizes protein and chlorophyll
- promotes root formation on a shoot
- low concentration encourages leaf senescence
- normal concentration increases cell enlargement in leaves

- stimulates lateral bud development

Ethylene (ethyne/C₂H₄)

- accelerates ripening in fruits
- encourages fruit fall/leaf fall
- induces thickening in stem/inhibits stem elongation
- promotes flowering (in pineapples)
- promotes germination in certain seeds

Absciscic acid (ABA) abscisin hormone/dormin)

- causes bud dormancy
- encourages fruit/leaf fall
- high concentration causes closing of stomata
- causes seed dormancy
- inhibits cell elongation

Traumatin

- heals wounds by callous formation

Florigen

- promotes flowering

ii) State the applications of plant hormones in agriculture

- induce root growth in stem cuttings
- selective weed killers
- encourage sprouting of lateral buds
- breaking seed dormancy
- induce parthenocarpy
- accelerate ripening of fruits
- promote flowering
- cause dormancy

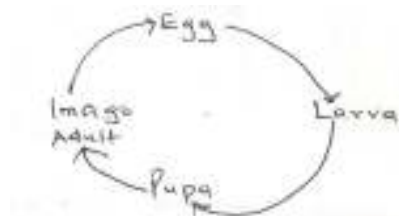
iii) Explain apical dominance



- a phenomenon whereby production of auxins by a growing apical bud of a shoot inhibits growth of lateral buds
- this inhibition is due to high concentration of auxins (indoleacetic acid/IAA) in apical bud
- removal of terminal/apical bud causes development and sprouting of several buds which later develop into branches
- applied in pruning coffee, tea and hedges
- this leads to more yield

example is egg larva pupa adult (imago)

occurs in animals such as butterfly and bee



iv) Describe the role of hormones in the growth and development of animal

somatotrophin (growth hormones)

- from anterior pituitary
- promotes cell division
- overproduction causes gigantism
- underproduction causes dwarfism

Thyroxine

- promotes growth and metamorphosis
- underproduction leads to a child becoming a cretin (mentally retarded)

Androgens

- in males
- growth of male reproductive organs

Oestrogen

- in females
- growth of female reproductive organs

Ecdysone

- in arthropods
- moulting (ecdysis)

t) i) What is metamorphosis?

- change in form during which there are changes in structure and function in body of organism
- prepares organism for life in a different habitat

ii) Explain complete metamorphosis

radical changes in the body during the life cycle of an organism

called holometabolous development

iii) What is the significance of each of the four stages in complete metamorphosis?

Larval stage

- feeding takes place
- larva is quite different from adult
- larva sheds its cuticle (exoskeleton) several times to emerge as pupa
- dispersal stage avoids overcrowding

Pupa

- enclosed in a case called puparium (cocoon)
- no feeding
- organ formation takes place

Adult

- emerges from puparium
- reproductive stage of the life cycle

iv) Describe incomplete metamorphosis

- called hemimetabolous development
- changes are gradual
- eggs develop into nymphs which develop into adults
- nymph resembles adult but are sexually immature



- a nymph moults several times as some parts develop before it becomes an adult
- stage of development between one moult and another is called instar
- occurs in insects such as locust and cockroach

Egg → Nymph → Adult

v) Name the hormones that control metamorphosis in insects

- brain hormone responsible for moulting because it stimulates production of ecdyson (moulting hormone)
- ecdysone(moulting hormone) causes moulting
- juvenile hormone causes moulting in larvae

vi) State the advantages of metamorphosis in the life of insects

- the adult and larvae exploit different niches
- do not compete for food
- pupa can survive adverse pupa can survive adverse conditions eg-feeding stage
- dispersal prevents overcrowding